

Notes on Semigroups and Related Structures

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1 Elementary Concepts

1.1 Sets and Relations

1.1.1 Relations

Definition 1.1.1 (Binary relation). Let X be a set. Every subset ρ of the Cartesian product $X \times X$ is called a *(binary) relation* on X . If $(x, y) \in \rho$, we also write $x\rho y$, where $x, y \in X$.

Definition 1.1.2 (Domain and range). For a relation $\rho \subseteq X \times X$, the *domain* and *range* of R are defined by

$$\text{dom } \rho = \{x \in X : \exists y \in X \text{ with } (x, y) \in R\},$$

and

$$\text{ran } \rho = \{y \in X : \exists x \in X \text{ with } (x, y) \in R\}.$$

The word *image* is also used sometimes as a synonym of range.

Notation 1.1.3. The set of all binary relations on X is denoted by $\mathfrak{B}(X)$.

Remark 1.1.4. If $\rho, \tau \in \mathfrak{B}(X)$, then $\rho \subseteq \tau$ means that $x\rho y \Rightarrow x\tau y$ for all $x, y \in X$. We shall often write $\rho \leq \tau$ for $\rho \subseteq \tau$.

Definition 1.1.5 (Complement). The complement of a relation $\rho \in \mathfrak{B}(X)$ is the relation $\bar{\rho} \in \mathfrak{B}(X)$ defined by

$$(x, y) \in \bar{\rho} \iff (x, y) \notin \rho.$$

Definition 1.1.6 (Properties of a relation). A relation $\rho \in \mathfrak{B}(X)$ is called

1. *reflexive*, if $x\rho x$ for all $x \in X$;
2. *symmetric*, if $x\rho y$ implies $y\rho x$;
3. *antisymmetric*, if $x\rho y$ and $y\rho x$ together imply $x = y$;
4. *transitive*, if $x\rho y$ and $y\rho z$ imply $x\rho z$;
5. *right unique*, if ρy and $x\rho z$ imply $y = z$;
6. *left unique*, if $x\rho y$ and $z\rho y$ imply $x = z$;
7. *right total*, if for every $y \in X$ there exists $x \in X$ such that $x\rho y$;
8. *left total*, if for every $x \in X$ there exists $y \in X$ such that $x\rho y$.

The last four notions also apply to relations $\rho \subseteq X \times Y$ for two possibly different sets X and Y .

1.1.2 Mappings

Definition 1.1.7 (Mapping). Let X and Y be sets. A mapping $\varphi : X \rightarrow Y$ is a relation $\varphi \subseteq X \times Y$ which is right unique and left total; that is, for every $x \in X$ there exists exactly one $y \in Y$ such that $(x, y) \in \varphi$.

Remark 1.1.8. As usual, we write $\varphi(x) = y$. Sometimes we also write $x\varphi = y$. The set of all mappings from X to Y is denoted by $\text{Map}(X, Y)$ or Y^X . We also use $\mathcal{T}(X)$ for $\text{Map}(X, X)$. The elements of $\mathcal{T}(X)$ are called *transformations* of X .

Definition 1.1.9 (Injective, surjective, bijective). A mapping $\varphi : X \rightarrow Y$ is called

1. *injective*, if $\varphi(x) = \varphi(x') \Rightarrow x = x'$ for all $x, x' \in X$;
2. *surjective*, if for every $y \in Y$ there exists $x \in X$ such that $\varphi(x) = y$;
3. *bijective*, if it is both injective and surjective.

Equivalently, injectivity means that the relation is left unique, and surjectivity means that the relation is right total.

Remark 1.1.10. The subset of all bijective transformations in $\mathcal{T}(X)$ is denoted by $\mathcal{S}(X)$, and its elements are called *permutations* or *bijections* of X .

Definition 1.1.11 (Identity map). The identity map $\text{id}_X : X \rightarrow X$ is defined by

$$\text{id}_X(x) = x \quad \forall x \in X.$$

Definition 1.1.12 (Constant mapping). For $y \in Y$, the constant mapping $c_y : X \rightarrow Y$ is defined by

$$c_y(x) = y \quad \forall x \in X.$$

Definition 1.1.13 (Restriction). If $\varphi : X \rightarrow Y$ is a mapping and $A \subseteq X$, then the restriction of φ to A is defined by $\varphi|_A : A \rightarrow Y$ such that

$$\varphi|_A(a) = \varphi(a) \quad \forall a \in A.$$

Conversely, we say that φ extends $\varphi|_A$.

Definition 1.1.14 (Partial mapping). A partial mapping $\varphi : X \rightarrow Y$ is defined to be a right unique relation; that is, $\varphi : \text{dom } \varphi \rightarrow Y$ is a mapping, where $\text{dom } \varphi \subseteq X$. The set $\text{dom } \varphi$ is called the domain of φ . The set of all partial mappings on X (from X to X) is denoted by $\mathcal{PT}(X)$.

Remark 1.1.15. The mapping $\varphi : \text{dom } \varphi \rightarrow \text{im } \varphi$ with $\text{dom } \varphi = \text{im } \varphi = \emptyset$ is called the empty mapping.

1.1.3 Order Relations

Definition 1.1.16 (Partial order). A reflexive, antisymmetric, and transitive relation α on a set X is called a *partial ordering* on X , or simply an ordering of X . One writes $x \leq y$ for $(x, y) \in \alpha$, and the pair $(X, \leq)$ is called a *partially ordered set* or a *poset*.

Definition 1.1.17 (Total order). If in addition, a partial ordering has the property that for every $x, y \in X$, either $x \leq y$ or $y \leq x$. In this case, $\leq$ is called a *total order*. Then $(X, \leq)$ is also called a *linearly ordered set* or a *chain*.

Definition 1.1.18 (Maximal and minimal elements). Let X be a poset. An element $w \in X$ is called a *maximal* (respectively, *minimal*) element of $(X, \leq)$ if $w \leq u$ (respectively, $u \leq w$) implies $u = w$ for every $u \in X$.

Definition 1.1.19 (Minimal condition and well-ordering). A poset $(X, \leq)$ satisfies the *minimal condition* if every non-empty subset of X has a minimal element. In that case, we also say that $(X, \leq)$ is *well ordered*.

Definition 1.1.20 (Lower bound and meet). Let Y be a non-empty subset of a poset X . An element $c \in X$ is called a *lower bound* for Y if $c \leq y$ for all $y \in Y$. An element $u \in X$ is called the *greatest lower bound* or *meet* of Y if u is a lower bound for Y and $z \leq u$ for every lower bound z of Y . The greatest lower bound of Y is denoted by $\bigwedge y : y \in Y$. If $Y = \{a, b\}$, we also write $a \wedge b$.

Definition 1.1.21 (Lower semilattice). If $(X, \leq)$ is such that $a \wedge b$ exists for every pair $a, b \in X$, then $(X, \leq)$ is called a *lower semilattice*. If, in addition, $\bigwedge \{y : y \in Y\}$ exists for every non-empty subset $Y \subseteq X$, then $(X, \leq)$ is called a *complete lower semilattice*.

Definition 1.1.22 (Upper bound and join). Similarly, for a non-empty subset $Y \subseteq X$, an element $c \in X$ is called an *upper bound* for Y if $y \leq c$ for all $y \in Y$. An element $v \in X$ is called the *least upper bound* or *join* of Y if v is an upper bound for Y and $v \leq z$ for every upper bound z of Y . The least upper bound of Y is denoted by $\bigvee \{y : y \in Y\}$. If $Y = \{a, b\}$, we also write $a \vee b$.

Definition 1.1.23 (Upper semilattice). If $(X, \leq)$ is such that $a \vee b$ exists for every pair $a, b \in X$, then $(X, \leq)$ is called an *upper semilattice*. If, in addition, $\bigvee y : y \in Y$ exists for every non-empty subset $Y \subseteq X$, then $(X, \leq)$ is called a *complete upper semilattice*.

Definition 1.1.24 (Lattice). If a poset is both a (complete) lower semilattice and a (complete) upper semilattice then it is called a (complete) lattice. In this case we use the notation $(X, \leq, \wedge, \vee)$.

Remark 1.1.25. The set $\mathfrak{B}(X)$ of all binary relations on a set X is a poset under inclusion; that is, $\rho \leq \tau$ if and only if $\rho \subseteq \tau$, for $\rho, \tau \in \mathfrak{B}(X)$. Moreover, it is a lattice with

$$\rho \wedge \tau = \rho \cap \tau \quad \text{and} \quad \rho \vee \tau = \rho \cup \tau,$$

which is called the lattice of relations on X and is denoted by $(\mathfrak{B}(X), \leq, \wedge, \vee)$.

Lemma 1.1.1 (Zorn's lemma). If $(X, \leq)$ is a non-empty poset such that every chain of elements has an upper bound in X , then X has at least one maximal element.

Theorem 1.1.2 (Zermelo's theorem on well-ordering). Every non-empty set can be well ordered.

1.1.4 Equivalence relations and partitions

Definition 1.1.26. An *equivalence* or *equivalence relation* on X is a reflexive, symmetric and transitive relation on X .

Special equivalence relations on X are $\text{id}_X = \{(x, x) \mid x \in X\}$, the *diagonal relation* on X , also denoted by Δ_X , and the *total relation* $X \times X$, also denoted by ∇_X .

Definition 1.1.27. Let $K \subseteq X$ be a subset. Define the *Rees equivalence* ρ_K by setting

$$x \rho_K x' \iff x, x' \in K \text{ or } x = x'.$$

We denote the set of all equivalences on a set X by $\mathcal{E}(X)$. As a subset of $\mathfrak{B}(X)$ it is partially ordered by inclusion.

We call the intersection of all equivalences containing the relation $\rho \in \mathcal{B}(X)$ the *equivalence on X generated by ρ* or the *equivalence closure* of ρ , and denote it by ρ^e .

If $\{\rho_i \in \mathcal{E}(X) \mid i \in I\}$ is a non-empty set of equivalences on X , then $\bigcap_{i \in I} \rho_i$ is again an equivalence on X .

If $\rho \in \mathcal{B}(X)$, then the intersection of all equivalences containing ρ is the unique minimal equivalence on X containing ρ .

In fact $\mathcal{E}(X)$ is a lattice. If $\rho, \sigma \in \mathcal{E}(X)$, then their greatest lower bound $\rho \wedge \sigma$ is $\rho \cap \sigma$ and their least upper bound $\rho \vee \sigma$ is $(\rho \cup \sigma)^e$.

Definition 1.1.28. For $\rho \in \mathcal{E}(X)$ and $A \subseteq X$, set

$$\rho(A) := \{b \in X \mid \text{there exists } a \in A \text{ with } (a, b) \in \rho\}.$$

For $A = \{a\}$ we write instead of $\rho(a)$ also $[a]_\rho$ or just $[a]$, and call this set the ρ -class of a or the class of a (with respect to ρ).

Let $\rho, \sigma \in \mathcal{E}(X)$. Then $\rho \subseteq \sigma$ if and only if every σ -class is a union of ρ -classes.

Definition 1.1.29. A family of subsets X_i , $i \in I$, of a set X is called a *partition* of X if the subsets X_i are pairwise disjoint, non-empty, and their union is X .

If ρ is an equivalence relation on X , then the set of equivalence classes

$$\pi(\rho) = \{[x]_\rho \mid x \in X\}$$

is the partition of X corresponding to ρ .

On the other hand, if $\pi = \{X_i \mid i \in I\}$ is a partition of X into subsets X_i , then the relation

$$\rho(\pi) = \{(x, y) \in X \times X \mid \text{there exists } i \in I \text{ such that } x, y \in X_i\}$$

is an equivalence relation on X . Note that

$$\rho(\pi(\rho)) = \rho, \quad \pi(\rho(\pi)) = \pi.$$

The set of classes with respect to an equivalence relation ρ is called the *factor set* of X with respect to ρ and denoted by X/ρ . The mapping

$$\pi_\rho : X \rightarrow X/\rho, \quad x \mapsto [x]_\rho$$

is called the *canonical surjection* with respect to ρ .

For any $\rho \in \mathfrak{B}(X)$, let

$$\begin{aligned}\rho^R &= \bigcap \{\sigma \in B(X) \mid \rho \subseteq \sigma, \sigma \text{ is reflexive}\}, \\ \rho^S &= \bigcap \{\sigma \in B(X) \mid \rho \subseteq \sigma, \sigma \text{ is symmetric}\}.\end{aligned}$$

There is at least one element $\sigma \in \mathfrak{B}(X)$ satisfying the condition in each of the collections above, namely $\sigma = X \times X$. Furthermore, since every such σ contains ρ , the intersections ρ^R and ρ^S are both non-empty. It is easy to see that

- ρ^R , called the *reflexive closure* of ρ , is the smallest reflexive relation containing ρ ;
- ρ^S , called the *symmetric closure* of ρ , is the smallest symmetric relation containing ρ .

1.1.5 Inverse relation, restriction and composition

Let $\rho \subseteq X \times Y$. The *inverse* (*converse*) relation to ρ is

$$\rho^{-1} = \{(y, x) \in Y \times X \mid (x, y) \in \rho\}.$$

Similar as for mappings, we now define the *restriction* $\rho|_{X'}$ of a relation $\rho \in B(X)$ for a subset $X' \subseteq X$ by

$$\rho|_{X'} = \rho \cap (X' \times X').$$

The *composition* of relations $\rho, \sigma \in \mathfrak{B}(X)$ is defined as

$$\rho \circ_r \sigma = \{(x, z) \in X \times X \mid \text{there exists } y \in X \text{ with } x\sigma y \text{ and } y\rho z\},$$

if we start with the right relation σ , and analogously for $\rho \subseteq X \times Y, \sigma \subseteq Y \times Z$ as

$$\rho \circ_r \sigma = \{(x, z) \in X \times Z \mid \text{there exists } y \in Y \text{ with } x\sigma y \text{ and } y\rho z\}.$$

If we start with the left relation ρ , we define

$$\rho \circ_l \sigma = \{(x, z) \in X \times X \mid \text{there exists } y \in X \text{ with } x\rho y \text{ and } y\sigma z\},$$

for $\rho, \sigma \in B(X)$, and analogously for $\rho \subseteq X \times Y, \sigma \subseteq Y \times Z$ as

$$\rho \circ_l \sigma = \{(x, z) \in X \times Z \mid \text{there exists } y \in Y \text{ with } x\rho y \text{ and } y\sigma z\}.$$

Usually, for convenience, we will denote either $\circ_r$ or $\circ_l$ by $\circ$. We will be careful in any situation to indicate the use of either convention.

If $\varphi \in \text{Map}(X, Y)$ and $\psi \in \text{Map}(Y, Z)$, then in the case when mapping symbols are written to the left of the elements on which they act, by $\psi\varphi$ we mean $\psi \circ_r \varphi$, i.e.

$$(\psi\varphi)(x) = \psi(\varphi(x)) \quad \forall x \in X.$$

In the case when mapping symbols are written to the right of the elements on which they act, by $x\varphi\psi$ we mean $\varphi \circ_l \psi$, i.e.

$$(x\varphi)\psi = (x\varphi)\psi \quad \forall x \in X.$$

1.1.6 Transitive and equivalence closure

It is clear that the operation $\circ$ on $\mathfrak{B}(X)$ is associative and thus $(\mathfrak{B}(X), \circ)$ will be one of the examples of monoids (definition in a further section) with the identity element id_X . Sometimes we write $\rho\sigma$ for $\rho \circ \sigma$. We define powers of a relation $\rho \in \mathfrak{B}(X)$ as follows:

$$\rho^0 = id_X, \quad \rho^i = \rho \circ \rho \circ \dots \circ \rho, \text{ } i \text{ times, } i \in \mathbb{N}.$$

Definition 1.1.30. The *transitive closure* of a relation $\rho \in \mathfrak{B}(X)$ is the relation

$$\rho^\infty = \bigcup_{n=1}^{\infty} \rho^n \in \mathfrak{B}(X).$$

Lemma 1.1.3. If $\rho \in \mathfrak{B}(X)$, then ρ^∞ is the smallest transitive relation in $(\mathfrak{B}(X), \leq)$ containing ρ .

Proof. If $(x, y), (y, z) \in \rho^\infty$, then there exist $m, n \in \mathbb{N}$ such that $(x, y) \in \rho^m, (y, z) \in \rho^n$, and

$$(x, z) \in \rho^m \circ \rho^n = \rho^{m+n} \subseteq \rho^\infty.$$

Thus ρ^∞ is a transitive relation and $\rho \subseteq \rho^\infty$. If now σ is a transitive relation on X and $\rho \subseteq \sigma$, then

$$\rho^2 = \rho \circ \rho \subseteq \sigma \circ \sigma \subseteq \sigma.$$

By induction it follows easily that $\rho^n \subseteq \sigma$ for all $n \in \mathbb{N}$, i.e. $\rho^\infty \subseteq \sigma$. □

Theorem 1.1.4. If $\rho \in \mathfrak{B}(X)$, then for the equivalence closure ρ^e we get

$$\rho^e = (\rho \cup \rho^{-1} \cup id_X)^\infty.$$

Proof. It is clear that $\tau = (\rho \cup \rho^{-1} \cup id_X)^\infty$ contains ρ and is transitive and reflexive. The relation $\nu = \rho \cup \rho^{-1} \cup id_X$ is symmetric and therefore $\nu^n = (\nu^{-1})^n = (\nu^n)^{-1}$. Thus, if $(x, y) \in \tau$, then $(x, y) \in \nu^n$ for some $n \in \mathbb{N}$ and therefore $(y, x) \in \nu^n$. Thus $(y, x) \in \tau$, i.e. τ is symmetric. This means that $\tau \in \mathcal{E}(X)$.

If $\sigma \in \mathcal{E}(X)$ and $\rho \subseteq \sigma$, then $id_X \subseteq \sigma, \rho^{-1} \subseteq \sigma^{-1} = \sigma$, and therefore $\nu = \rho \cup \rho^{-1} \cup id_X \subseteq \sigma$. Since $\nu \circ \nu \subseteq \sigma \circ \sigma \subseteq \sigma$, we get $\nu^n \subseteq \sigma$ for all $n \in \mathbb{N}$, and hence $\tau \subseteq \sigma$. □

Corollary 1.1.5. If $\rho \in \mathfrak{B}(X)$, then $(x, y) \in \rho^e$ if and only if $x = y$ or for some $n \in \mathbb{N}$ there exists a sequence of elements $x_0, x_1, x_2, \dots, x_n$ in X such that $x_0 = x, x_n = y$, and for every $i \in \{1, \dots, n\}$ either $(x_{i-1}, x_i) \in \rho$ or $(x_i, x_{i-1}) \in \rho$.

In particular, if ρ and σ are equivalence relations on a set X , then in $\mathcal{E}(X)$ their join $\rho \vee \sigma$ is the relation defined by

$$x(\rho \vee \sigma)y \iff \text{there exist } x_0, x_1, \dots, x_n \in X \text{ such that } x = x_0, y = x_n$$

and

$$(x_{i-1}, x_i) \in \tau_i, \quad \tau_i \in \{\rho, \sigma\}, \quad i \in \{1, \dots, n\}.$$

Proposition 1.1.6. For any $\rho \in \mathfrak{B}(X)$,

1. $\rho^R = \rho \cup id_X$;
2. $\rho^S = \rho \cup \rho^{-1}$;
3. $(\rho^R)^S = (\rho^S)^R = \rho \cup \rho^{-1} \cup id_X$;
4. $\rho^e = \left((\rho^R)^S \right)^\infty$.

1.1.7 Kernel equivalence and Homomorphism Theorem for sets

Definition 1.1.31. Let $\varphi : X \rightarrow Y$ be a mapping. The relation

$$\ker \varphi = \{(x, x') \in X \times X \mid \varphi(x) = \varphi(x')\}$$

is an equivalence on X which is called the *kernel equivalence* or the *kernel* of φ . It follows from the definition of the composition of relations that

$$\ker \varphi = \varphi^{-1} \circ \varphi = \varphi^{-1} \circ_r \varphi.$$

Theorem 1.1.7 (Homomorphism Theorem for sets). *Let $\varphi : X \rightarrow Y$ be a mapping and let ρ be an equivalence on X such that $x\rho x'$ implies $\varphi(x) = \varphi(x')$, i.e. $\rho \leq \ker \varphi$. Then $\varphi' : X/\rho \rightarrow Y$, with $\varphi'([x]_\rho) := \varphi(x)$, $x \in X$, is the unique mapping such that the following diagram is commutative:*

$$\begin{array}{ccc} X & \xrightarrow{\varphi} & Y \\ \pi_\rho \downarrow & \nearrow \varphi' & \\ X/\rho & & \end{array}$$

i.e. such that $\varphi' \pi_\rho = \varphi$, where π_ρ denotes the canonical surjection.

If $\rho = \ker \varphi$, then φ' is injective, and if φ is surjective, then φ' is surjective.

Proof. Define $\varphi' : X/\rho \rightarrow Y$ by $\varphi'([x]_\rho) = \varphi(x)$ for any $x \in X$. Suppose $[x]_\rho = [x']_\rho$ for $x, x' \in X$. Then $(x, x') \in \rho$ and thus $\varphi(x) = \varphi(x')$. Hence φ' is well-defined.

Suppose $\varphi'([x]_\rho) = \varphi'([x']_\rho)$. Then $\varphi(x) = \varphi(x')$ and thus $(x, x') \in \ker \varphi$.

If $\rho = \ker \varphi$, then $[x]_\rho = [x']_\rho$ and thus φ' is injective. □

1.1.8 Congruences

Definition 1.1.32. A relation ρ on X is called

- *left compatible* if for all $x, y, z \in X$,

$$(x, y) \in \rho \implies (zx, zy) \in \rho;$$

- *right compatible* if for all $x, y, z \in X$,

$$(x, y) \in \rho \implies (xz, yz) \in \rho;$$

- *compatible* if for all $x, y, s, t \in X$,

$$(x, y) \in \rho \text{ and } (s, t) \in \rho \implies (xs, yt) \in \rho.$$

A left compatible equivalence relation is called a *left congruence*; a right compatible equivalence relation is called a *right congruence*; and a compatible equivalence relation is called a *congruence*.

Proposition 1.1.8. *A relation ρ on X is a congruence if and only if it is both a left and a right congruence.*

1 Elementary Concepts

Proof. Suppose ρ is both a left and a right congruence. Let $x, y, z, t \in X$ be such that $(x, y) \in \rho$ and $(z, t) \in \rho$. Since ρ is a right congruence, $(xz, yt) \in \rho$, and since ρ is a left congruence, $(yz, yt) \in \rho$. By transitivity of ρ , it follows that $(xz, yt) \in \rho$. Hence ρ is a congruence.

Conversely, suppose ρ is a congruence. Let $x, y \in X$ be such that $(x, y) \in \rho$, and let $z \in X$. Since ρ is reflexive, $(z, z) \in \rho$. By compatibility, $(zx, zy) \in \rho$. Thus ρ is left compatible. Similarly, $(xz, yz) \in \rho$, so ρ is right compatible. Hence ρ is both a left and a right congruence. $\square$

For any $\rho \in \mathfrak{B}(X)$, let

$$\rho^c = \bigcap \{ \sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is left and right compatible} \}.$$

Then ρ^c is the smallest left and right compatible relation containing ρ . Also, let

$$\rho^\# = \bigcap \{ \sigma \in \mathfrak{B}(X) \mid \rho \subseteq \sigma, \sigma \text{ is a congruence} \}.$$

Then ρ^c , called the congruence generated by ρ , is the smallest congruence containing ρ .

Proposition 1.1.9. For any $\rho, \sigma \in \mathfrak{B}(X)$,

1. $(\rho \cup \sigma)^c = \rho^c \cup \sigma^c$;
2. $(\rho^{-1})^c = (\rho^c)^{-1}$.

Proposition 1.1.10. For any $\rho \in \mathfrak{B}(X)$, we have

$$\rho^\# = (\rho^c)^e.$$

Proof. By definition, $(\rho^c)^e$ is an equivalence relation containing ρ^c , and hence containing ρ . Notice that

$$\begin{aligned} (\rho^c)^e &= (\rho^c \cup (\rho^c)^{-1} \cup id_X)^\infty \\ &= ((\rho \cup \rho^{-1} \cup id_X)^c)^\infty \\ &= \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \end{aligned}$$

Now, for all $x, y, z \in X$,

$$\begin{aligned} &(x, y) \in (\rho^c)^e \\ \implies &(x, y) \in \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &\exists n \in \mathbb{N} \text{ such that } (x, y) \in ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &\exists x_0, x_1, \dots, x_n \text{ such that } x = x_0 \text{ and } y = x_n \text{ and } (\forall i) (x_i, x_{i+1}) \in (\rho \cup \rho^{-1} \cup id_X)^c \\ \implies &zx = zx_0, zx_n = zy \text{ and } (\forall i) (zx_i, zx_{i+1}) \in (\rho \cup \rho^{-1} \cup id_X)^c \\ &\quad (\text{since } (\rho \cup \rho^{-1} \cup id_X)^c \text{ is left and right congruence}) \\ \implies &(zx, zy) \in ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &(zx, zy) \in \bigcup_{n=0}^\infty ((\rho \cup \rho^{-1} \cup id_X)^c)^n \\ \implies &(zx, zy) \in (\rho^c)^e. \end{aligned}$$

Therefore, $(\rho^c)^e$ is left compatible. Similarly, $(\rho^c)^e$ is right compatible. Hence $(\rho^c)^e$ is a congruence containing ρ .

Finally, let τ be any congruence containing ρ . Then τ is left and right compatible, so $\rho^c \subseteq \tau$. Since τ is an equivalence relation, it must contain $(\rho^c)^e$. Hence $(\rho^c)^e \subseteq \tau$. Therefore, $(\rho^c)^e$ is the smallest congruence containing ρ , and therefore $\rho^\# = (\rho^c)^e$. Consequently, $\rho^\# = (\rho^c)^e$ $\square$

We denote by $\mathcal{C}(X)$ the set of all congruences on X . Then $\mathcal{C}(X)$, as a subset of $\mathcal{E}(X)$ and $\mathcal{B}(X)$, is partially ordered by inclusion.

In fact, $\mathcal{C}(X)$ forms a lattice. For $\rho, \sigma \in \mathcal{C}(X)$, we have

$$\rho \wedge \sigma = \rho \cap \sigma \quad \text{and} \quad \rho \vee \sigma = (\rho \cup \sigma)^\#.$$

Also,

$$\begin{aligned} (\rho \cup \sigma)^\# &= ((\rho \cup \sigma)^c)^e \\ &= (\rho^c \cup \sigma^c)^e \\ &= (\rho \cup \sigma)^e \end{aligned}$$

so there is no ambiguity in writing $\rho \vee \sigma$ whether one means the join $(\rho \vee \sigma)^e$ in the lattice $\mathcal{E}(X)$ or the join $(\rho \vee \sigma)^\#$ in the lattice $\mathcal{C}(X)$.

Proposition 1.1.11. *Let $\rho, \sigma \in \mathcal{C}(X)$. Then*

$$\rho \vee \sigma = (\rho \circ \sigma)^\infty.$$

Proof. Let $\rho, \sigma \in \mathcal{C}(X)$. Since $\rho \vee \sigma$ contains both ρ and σ , it follows that

$$\rho \circ \sigma \subseteq (\rho \cup \sigma) \circ (\rho \cup \sigma) = (\rho \vee \sigma)^2.$$

More generally, $(\rho \circ \sigma)^n \subseteq (\rho \cup \sigma)^{2n}$. Thus,

$$(\rho \circ \sigma)^\infty = \bigcup_{n=0}^{\infty} (\rho \circ \sigma)^n \subseteq \bigcup_{n=0}^{\infty} (\rho \cup \sigma)^{2n} = (\rho \cup \sigma)^\infty.$$

On the other hand, $\rho \circ \sigma \supseteq \rho \circ id_X = \rho$ (since σ is reflexive), and $\rho \circ \sigma \supseteq id_X \circ \sigma = \sigma$ (since ρ is reflexive). Thus, $\rho \cup \sigma \subseteq \rho \circ \sigma$. Hence, $(\rho \cup \sigma)^\infty \subseteq (\rho \circ \sigma)^\infty$.

Consequently, $(\rho \cup \sigma)^\infty = (\rho \circ \sigma)^\infty$.

Then,

$$\begin{aligned} \rho \vee \sigma &= (\rho \cup \sigma)^e = \left(((\rho \cup \sigma)^R)^S \right)^\infty \\ &= ((\rho \cup \sigma) \cup (\rho \cup \sigma)^{-1} \cup id_X)^\infty \\ &= (\rho \cup \sigma \cup \rho^{-1} \cup \sigma^{-1} \cup id_X)^\infty \\ &= (\rho \cup \sigma \cup id_X)^\infty \\ &= (\rho \cup \sigma)^\infty = (\rho \circ \sigma)^\infty. \end{aligned}$$

This completes the proof. □

Proposition 1.1.12. *Let $\rho, \sigma \in \mathcal{C}(X)$. If $\rho \circ \sigma = \sigma \circ \rho$, then*

$$\rho \vee \sigma = \rho \circ \sigma.$$

Proof. Suppose $\rho \circ \sigma = \sigma \circ \rho$. Then

$$(\rho \circ \sigma)^2 = \rho \circ \sigma \circ \rho \circ \sigma = \rho \circ \rho \circ \sigma \circ \sigma = \rho \circ \sigma,$$

since ρ and σ are transitive.

Hence, $(\rho \circ \sigma)^n = \rho \circ \sigma$ for all $n \in \mathbb{N}$, and therefore

$$\rho \circ \sigma = (\rho \circ \sigma)^\infty = \bigcup_{n \geq 0} (\rho \circ \sigma)^n = \rho \circ \sigma.$$

So $\rho \vee \sigma = \rho \circ \sigma$. □

Lemma 1.1.13. *If $\rho = \cap_i \rho_i$ (an intersection of congruences, or even just equivalence relations on a set X), then for any $x \in X$*

$$[x]_\rho = \cap_i [x]_{\rho_i}.$$

Proof. ; For any $y \in X$

$$x \in [x]_\rho \iff (x, y) \in \rho \iff (x, y) \in \rho_i \quad \forall i \iff y \in [x]_{\rho_i} \quad \forall i \iff y \in \cap_i [x]_{\rho_i}$$

□

1.2 Introductory Algebraic structures

1.2.1 Groupoids, Identities, and Zeros

Definition 1.2.1. A *groupoid* is a non-empty set A together with a binary operation

$$A \times A \rightarrow A, \quad (a, a') \mapsto aa'.$$

The operation is usually called the *multiplication* of the groupoid.

Remark 1.2.2. Let A be a groupoid. For non-empty subsets $B, C \subseteq A$, define

$$BC = \{bc \mid b \in B, c \in C\}.$$

We write $B^2 = BB$, and similarly for higher products.

Definition 1.2.3. Let S be a groupoid. An element $e \in S$ is called

- a *left identity* if $es = s$ for all $s \in S$;
- a *right identity* if $se = s$ for all $s \in S$;
- an *identity* if $es = s = se$ for all $s \in S$.

An identity is also called a *neutral element*.

Proposition 1.2.1. *If a groupoid S has both a left and a right identity, then they coincide. Hence a groupoid has at most one identity.*

Proof. Let e' be a left identity and e'' a right identity. Then

$$e'' = e'e'' = e'.$$

□

We denote the identity (when it exists) by 1 or more precisely by 1_S .

Definition 1.2.4. Let S be a groupoid with identity.

- $s \in S$ is *left invertible* if there exists $t \in S$ such that $ts = 1$.
- s is *right invertible* if there exists $t \in S$ such that $st = 1$.
- s is *invertible* if there exists $t \in S$ satisfying $ts = st = 1$.

In third case, such an element t is called an inverse of s , and is denoted by s^{-1} whenever it is unique.

Definition 1.2.5. Let S be a groupoid. An element $z \in S$ is called

- a *left zero* if $zs = z$ for every $s \in S$;
- a *right zero* if $sz = z$ for every $s \in S$;
- a *zero* if $zs = z = sz$ for every $s \in S$.

A zero is usually denoted by 0.

Exercise 1.2.2. Show that if a groupoid has both a left zero and a right zero, then they coincide. Hence a groupoid has at most one zero.

1.2.2 Semigroups, monoids and groups

Definition 1.2.6. A *semigroup* is a groupoid whose multiplication is associative. A semigroup with an identity element is called a *monoid*.

Proposition 1.2.3. If an element of a monoid has both a left and a right inverse, then they coincide.

Proof. Suppose $ts = 1$ and $st' = 1$. Then

$$t' = (ts)t' = t(st') = t. \quad \square$$

Corollary 1.2.4. Every element of a monoid has at most one inverse.

Definition 1.2.7. A *group* G is a monoid in which every element is invertible.

For any $a, b \in G$,

$$(ab)^{-1} = b^{-1}a^{-1}, \quad (a^{-1})^{-1} = a.$$

Exercise 1.2.5. Suppose S is a semigroup with an element e satisfying $es = s$ for every $s \in S$, and suppose every $s \in S$ has an element $t \in S$ such that $ts = e$. Show that S is a group.

Remark 1.2.8 (Free semigroup and free monoid). Let X be a non-empty set, called an *alphabet*. Define

$$X^+ = \{x_1x_2 \cdots x_n \mid n \in \mathbb{N}, x_i \in X, 1 \leq i \leq n\},$$

the set of all finite non-empty words over X .

Two words

$$x_1x_2 \cdots x_n, \quad y_1y_2 \cdots y_m$$

are equal if and only if

$$n = m \quad \text{and} \quad x_i = y_i \quad \text{for all } 1 \leq i \leq n.$$

Define the multiplication (concatenation) of words by

$$(x_1x_2 \cdots x_n)(y_1y_2 \cdots y_m) = x_1x_2 \cdots x_ny_1y_2 \cdots y_m.$$

With this operation, X^+ forms a semigroup, called the *free semigroup* on X , or the *free semigroup with basis X* .

Now let

$$X^* = X^+ \cup \{\varepsilon\},$$

where ε denotes the empty word. Then X^* is a monoid under concatenation, with identity element ε . It is called the *free monoid* on X , or the *free monoid with basis X* .

1.2.3 Examples

Example 1.2.9 (Groupoids). 1. $(\mathbb{Z}, -)$.

2. $(\mathbb{Q} \setminus \{0\}, /)$.

3. $(\mathbb{N}, \uparrow)$, where $n \uparrow m = n^m$.

4. The vector product $(\mathbb{R}^3, \times)$.

Example 1.2.10 (Semigroups and monoids). 1. If $(R, +, \cdot)$ is a ring, then $(R, +)$ is a group and $(R, \cdot)$ is a semigroup.

2. $(\mathbb{N}, \cdot)$ is a monoid.

3. $(\mathbb{N}, +)$ is a semigroup.

4. $(\mathcal{T}(X), \circ)$, the full transformation monoid.

5. $(\mathcal{PT}(X), \circ)$, the monoid of partial transformations.

6. $(\mathfrak{B}(X), \circ)$, the semigroup of binary relations.

7. The free semigroup X^+ .

8. The free monoid X^* .

9. $(\mathbb{Z}, \gcd)$.

10. **The bicyclic monoid:**

Let $B = \mathbb{N}_0 \times \mathbb{N}_0$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. Define a binary operation on B by

$$(m, n)(p, q) = (m - n + \max\{n, p\}, q - p + \max\{n, p\}).$$

Equivalently,

$$(m, n)(p, q) = \begin{cases} (m - n + p, q), & \text{if } p \geq n, \\ (m, n - p + q), & \text{if } p \leq n. \end{cases}$$

Then $(B, \cdot)$ is a monoid, called the *bicyclic monoid*, with identity element $(0, 0)$.

Example 1.2.11 (Groups). 1. $(\mathbb{Z}, +)$.

2. $(\mathbb{Q}, +)$.

3. $(\mathbb{Q} \setminus \{0\}, \cdot)$.

4. The residue class group $(\mathbb{Z}_r, +)$.

5. The symmetric group $S(X)$, or S_n when $|X| = n$.

2 The Semigroup

2.1 Semigroups and Monoids

Throughout this chapter, S will denote a semigroup with operation $\circ$. Formally, we write $(S, \circ)$ to indicate that we are considering the set S with the operation $\circ$, but we will only do this when we need to distinguish a particular operation. Unless we need to distinguish between different operations, we will often write xy instead of $x \circ y$ (where $x, y \in S$), and we will call the operation multiplication and the element xy (i.e. the result of applying the operation to x and y) the product of x and y .

Since the operation is associative, there is no ambiguity in the product $s_1 s_2 \cdots s_n$ (where each $s_i \in S$): the product is the same regardless of how we insert the brackets.

Any group is also a semigroup. The most familiar example of a semigroup that is not a group is the set of natural numbers $\mathbb{N}$ under the operation of addition. This is not a group since it contains no identity. (For our purposes, $\mathbb{N} = \{1, 2, 3, \dots\}$ does not include 0.)

Example 2.1.1. The trivial semigroup contains only one element e , with multiplication obviously defined by $ee = e$. Since e is (trivially) an identity, this semigroup is also called the trivial monoid.

2.1.1 Basic Concepts

Definition 2.1.2. A null semigroup is a semigroup with a zero z in which the product of any two elements is z . It is easy to see that this multiplication is associative. Notice that we can define a null semigroup on any set by choosing some element z and defining all products to be z .

Definition 2.1.3. If every element of S is a left zero (that is, $xy = x$ for all $x, y \in S$), then S is a left zero semigroup. If every element of S is a right zero (that is, $xy = y$ for all $x, y \in S$), then S is a right zero semigroup. We can define a left zero semigroup on any set X by defining the multiplication $xy = x$ for all $x, y \in X$; it is easy to see that this multiplication is associative.

If a semigroup does not contain identity (zero element), we can always add a new element to act as an identity (zero element). To see this, let 1 be a new element not in the semigroup S . Extend the multiplication on S to $S \cup \{1\}$ by $1x = x1 = x$ for all $x \in S$ and $11 = 1$. It is easy to prove that this extended multiplication is associative. Then $S \cup \{1\}$ is a monoid with identity 1 . Similarly, let 0 be a new element not in S and extend the multiplication on S to $S \cup \{0\}$ by $0x = x0 = 00 = 0$ for all $x \in S$. Again, this extended multiplication is associative. Then $S \cup \{0\}$ is a semigroup with zero 0 .

For any semigroup S , define

$$S^1 = \begin{cases} S, & \text{if } S \text{ has an identity,} \\ S \cup \{1\}, & \text{otherwise,} \end{cases} \quad S^0 = \begin{cases} S, & \text{if } S \text{ has a zero,} \\ S \cup \{0\}, & \text{otherwise.} \end{cases}$$

2 The Semigroup

The semigroups S^1 and S^0 are called, respectively, the monoid obtained by adjoining an identity to S if necessary and the semigroup obtained by adjoining a zero to S if necessary.

Throughout this chapter, mappings are written on the right and composed left to right. To clarify: let $\varphi : X \rightarrow Y$ and $\psi : Y \rightarrow Z$ be maps. The result of applying φ to an element x of X is denoted $x\varphi$. The composition of φ and ψ is denoted $\varphi\psi$ or simply $\varphi\psi$, and is a map from X to Z with $x(\varphi\psi) = (x\varphi)\psi$ for all $x \in X$.

Definition 2.1.4 (Cartesian Product). Let $X = \{X_i : i \in I\}$ be a collection of sets. Informally, the Cartesian product $\prod_{i \in I} X_i$ of the sets in X is the set of tuples with $|I|$ components, where, for each $i \in I$, the i -th component is an element of X_i . More formally, the Cartesian product $\prod_{i \in I} X_i$ is the set of mappings σ from I to $\bigcup_{i \in I} X_i$ such that $i\sigma \in X_i$ for each $i \in I$.

We think the map σ as a tuple with i -th component $i\sigma$. We will use both mapping notation and (especially when the index set I is finite) tuple notation for elements of Cartesian products. When $I = \{1, \dots, n\}$ is finite, we write $X_1 \times \dots \times X_n$ for $\prod_{i \in I} X_i$. When the sets X_i are all equal to a set X , that is, when we consider products of $|I|$ copies of the set X , we write X^I for $\prod_{i \in I} X_i$.

Definition 2.1.5. Let $S = \{S_i : i \in I\}$ be a collection of semigroups. The direct product of the semigroups in S is their Cartesian product $\prod_{i \in I} S_i$ with componentwise multiplication: $(i)(s)(i)(t) = (i)(s)(i)t$, or, using tuple notation, if $(\dots, s_i, \dots), (\dots, t_i, \dots) \in \prod_{i \in I} S_i$, then

$$(\dots, s_i, \dots)(\dots, t_i, \dots) = (\dots, s_i t_i, \dots).$$

It is easy to prove that this componentwise multiplication is associative, and so the direct product is itself a semigroup.

For $x \in S$ and $n \in \mathbb{N}$, define

$$x^n = \underbrace{x \cdots x}_{n \text{ times}}.$$

In general, x^n is only defined for positive n . If S is a monoid, define $x^0 = 1_S$. An element x of S is *idempotent* if $x^2 = x$. The set of idempotents of S is denoted $E(S)$.

For any subsets X and Y of S , define $XY = \{xy : x \in X, y \in Y\}$. Write xY for $\{x\}Y$ and XY for $X\{y\}$. Since multiplication in S is associative, so is this product of subsets: for subsets X, Y, Z of S , we have $X(YZ) = (XY)Z$. By analogy with the above, for $X \subseteq S$ and $n \in \mathbb{N}$, define

$$X^n = \underbrace{X \cdots X}_{n \text{ times}}.$$

Definition 2.1.6 (Nilpotent Semigroup). The semigroup S is nilpotent if it contains a zero and there exists some $n \in \mathbb{N}$ such that $S^n = \{0\}$.

Definition 2.1.7. The semigroup S is a nilsemigroup if it contains a zero and for every $x \in S$, there exists some $n \in \mathbb{N}$ such that $x^n = \{0\}$.

Definition 2.1.8 (Cancellative Semigroup). The semigroup S is left cancellative if

$$(\forall x, y, z \in S)(zx = zy \Rightarrow x = y);$$

right cancellative if

$$(\forall x, y, z \in S)(xz = yz \Rightarrow x = y);$$

and cancellative if it is both left and right cancellative.

Note 2.1.9. A non-trivial semigroup with zero cannot be cancellative.

Definition 2.1.10 (Commutativity). The semigroup S is commutative if $xy = yx$ for all $x, y \in S$. For instance, $(\mathbb{Z}, +)$ and $(\mathbb{Z}, \cdot)$ are both commutative.

A non-trivial left zero semigroup is not commutative.

2.1.2 Regular Semigroups

Let $x \in S$. If there is an element $y \in S$ such that $xyx = x$, then the element x is *regular*. Notice that in this case, $(xy)^2 = xyxy = x(yx)y = xy$ and $(yx)^2 = yxyx = y(xy)x = yx$ and hence xy and yx are idempotent elements of S .

Definition 2.1.11 (Regular Semigroup). If every element of S is regular, then S is a regular semigroup.

Now we define the inverse of an element of a semigroup S in general.

Definition 2.1.12. Let $x \in S$. An element $x' \in S$ is called as inverse of x if $x = xx'x$ and $x' = x'xx'$.

Notice that this is entirely different from the notion of left/right inverses above. We will never use *inverse* (on its own) to refer to a left or right inverse.

Proposition 2.1.1. Let $x \in S$. Then x has an inverse if and only if x is regular.

Proof. Obviously if x has an inverse, then it is regular. So suppose x is regular. Then there exists $y \in S$ such that $xyx = x$. Let $x' = yxy$. Then

$$xx'x = x(yxy)x = (xyx)yxyx = xyx = x$$

and

$$x'xx' = (yxy)x(yxy) = y(xy x)yxy = yxy = x',$$

so x' is an inverse of x . □

In the proof of Proposition 2.1.1, the element y might not be an inverse of x ; for example, let S be a semigroup with a zero and let $x = 0$ and $y \neq 0$.

An element x can have more than one inverse. The set of inverses of x is denoted $V(x)$. Notice also that a zero of a semigroup has an inverse, namely itself. In general, if $e \in S$ is idempotent, then $e^2 = e$ and so e is an inverse of itself. In particular, every idempotent is regular.

Example 2.1.13. (a) Let $U = \{0, \dots, k\}$ for some $k > 0$. Define $m\Delta n = \min\{m, n\}$. It is easy to see that Δ is associative, and so (U, Δ) is a semigroup. Notice that $0\Delta m = m\Delta 0 = 0$ and $k\Delta m = m\Delta k = m$ for all $m \in U$. Hence U has zero 0 and identity k . Furthermore, $m\Delta m = m$ for all $m \in U$, so every element of U is idempotent. Finally, $m\Delta n = n\Delta m$ for all $m, n \in U$, so U is commutative.

(b) Define a similar associative operation Δ on $\mathbb{N}_0 \cup \{0\} = \{0, 1, 2, \dots\}$ by

$$m\Delta n = \min\{m, n\}.$$

Then $(\mathbb{N}, \Delta)$ has a zero 0 but has no identity. It is commutative and all its elements are idempotents.

(c) Consider the set of all 2×2 integer matrices:

$$M_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z} \right\}.$$

2 The Semigroup

With the usual matrix multiplication, $M_2(\mathbb{Z})$ is a monoid with identity

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and zero

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

It is easy to see that $M_2(\mathbb{Z})$ is not commutative, that not all of its elements are idempotents. Since $M_2(\mathbb{Z})$ contains a zero, it is not cancellative.

- (d) Now let V be the set of all 2×2 integer matrices with non-zero determinant. Again, V is a monoid. Let $P, Q, R \in V$. Suppose $RP = RQ$. Since $\det R \neq 0$, the matrix R has an inverse $R^{-1} \in M_2(\mathbb{Q})$. Hence V is left-cancellative. Similarly, it is right-cancellative and therefore cancellative.

Definition 2.1.14. Let L be a left zero semigroup and R a right zero semigroup. Let $B = L \times R$. This semigroup is called as $|L| \times |R|$ rectangular band, or simply a *rectangular band*.

For $(\ell_1, r_1), (\ell_2, r_2) \in B$, we have

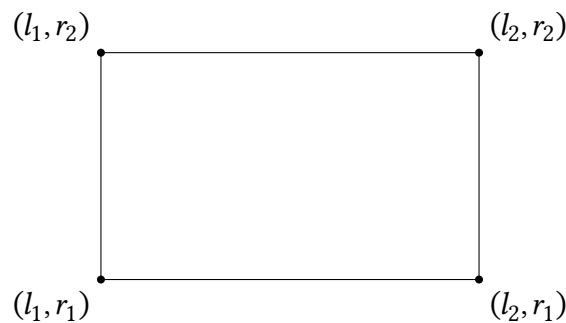
$$(\ell_1, r_1)(\ell_2, r_2) = (\ell_1, r_2),$$

since (in particular) ℓ_1 is a left zero and r_2 is a right zero. Thus every element of B is idempotent, since $(\ell, r)(\ell, r) = (\ell, r)$ for all $(\ell, r) \in B$. Furthermore, for any $(\ell_1, r_1), (\ell_2, r_2) \in B$, we have

$$(\ell_1, r_1)(\ell_2, r_2)(\ell_1, r_1) = (\ell_1 \ell_2 \ell_1, r_1 r_2 r_1) = (\ell_1, r_1),$$

$$(\ell_2, r_2)(\ell_1, r_1)(\ell_2, r_2) = (\ell_2 \ell_1 \ell_2, r_2 r_1 r_2) = (\ell_2, r_2).$$

Hence (ℓ_2, r_2) is an inverse of (ℓ_1, r_1) . Thus every element is an inverse of every element. The name *rectangular band* comes from the following geometric interpretation of the multiplication. To find the product $(\ell_1, r_1)(\ell_2, r_2)$, take the rectangle two of whose vertices have coordinates (ℓ_1, r_1) and (ℓ_2, r_2) . Then the product is the vertex vertically in line with (ℓ_1, r_1) and horizontally in line with (ℓ_2, r_2) , as shown in Figure 2.1.2.



2.1.3 Subsemigroups

Definition 2.1.15 (Subsemigroup). A non-empty subset T of S is a subsemigroup if it is closed under multiplication; that is, if $TT \subseteq T$. A proper subsemigroup is any subsemigroup except S itself. A submonoid is a subsemigroup that happens to be a monoid. A subgroup is a subsemigroup that happens to be a group.

Proposition 2.1.2. *The set of invertible elements of a monoid forms a subgroup.*

Proof. Let T be the set of invertible elements of a monoid M . Note that T is non-empty since $1 \in T$. Let $x, y \in T$. Then since x and y are invertible we have

$$y^{-1}x^{-1}xy = y^{-1}y = 1 \quad \text{and} \quad xy y^{-1}x^{-1} = xx^{-1} = 1.$$

Hence xy is invertible and so lies in T . Hence T is a subsemigroup of M . Furthermore, $1 \in T$ is also an identity for T , and so T is a submonoid of M . Finally, all elements of T are invertible, and so T is a subgroup of M . $\square$

Proposition 2.1.2 makes following definition justifiable:

Definition 2.1.16 (Group of units). The set of invertible elements of a monoid is called its group of units.

Lemma 2.1.3. *Let G be a subset of S . Then $gG = Gg = G$ for all $g \in G$ if and only if G is a subgroup of S .*

Proof. Notice first that if G is a subgroup and $g \in G$, then

$$G = gg^{-1}G \subseteq gG \subseteq G,$$

so $G = gG$. Similarly $G = Gg$.

For the converse, suppose that $gG = Gg = G$ for all $g \in G$. Let $g, h \in G$. Then $gh \in gG = G$, and so G is a subsemigroup.

Since $g \in G = Gg$, there exists $e \in G$ such that $eg = g$. Since $h \in G = gG$, we have $h = gx$ for some $x \in G$, and then

$$eh = egx = gx = h.$$

Thus e is a left identity for G . Similarly G contains a right identity f , and so $e = f$ is an identity for G by Proposition 1.2.1. So G is a submonoid with identity 1_G .

Finally, since $1_G \in gG = Gg$, the element g is right and left invertible, and its right and left inverses coincide by Proposition 1.2.3. Since $g \in G$ was arbitrary, G is a subgroup. $\square$

2.1.4 Generators

Let $\mathcal{T} = \{T_i : i \in I\}$ be a collection of subsemigroups of S . Then if their intersection $\bigcap T = \bigcap_{i \in I} T_i$ is non-empty, it is also a subsemigroup.

Definition 2.1.17 (Generating a subsemigroup). Let $X \subseteq S$, and let $\mathcal{T}$ be the collection of subsemigroups of S that contain X . The collection $\mathcal{T}$ has at least one member, namely the semigroup S itself, and every subsemigroup in $\mathcal{T}$ contains X , so $\bigcap T$ is non-empty and is thus a subsemigroup. Indeed, it is the smallest subsemigroup of S that contains X . This subsemigroup, denoted by $\langle X \rangle$, is called the subsemigroup generated by X .

If $X \subseteq S$ is such that $\langle X \rangle = S$, then X is a generating set for S and X generates S . If there is a finite generating set for S , then S is said to be finitely generated.

Proposition 2.1.4. *Let $X \subseteq S$. Then*

$$\langle X \rangle = \{x_1 x_2 \cdots x_n : n \in \mathbb{N}, x_i \in X\}.$$

2 The Semigroup

Proof. Let

$$U = \{x_1 x_2 \cdots x_n : n \in \mathbb{N}, x_i \in X\}.$$

Then U is closed under multiplication and so is a subsemigroup of S . Furthermore, $X \subseteq U$. Hence U must be one of the T_i in T , and so $\langle X \rangle \subseteq U$. Since $X \subseteq \langle X \rangle$ and $\langle X \rangle$ is closed under multiplication, we have $U \subseteq \langle X \rangle$. Therefore $\langle X \rangle = U$. $\square$

Given a subset X of a monoid M , we can also define the submonoid generated by X . This submonoid is the intersection of all submonoids of M containing whose identities are 1_M , that is, the collection of submonoids of M containing X and 1_M . The intersection of the submonoids in M is non-empty and thus a submonoid. This is the smallest submonoid of M with identity 1_M that contains X , denoted by $\text{Mon}(X)$.

Definition 2.1.18. This submonoid $\text{Mon}(X)$ is called the submonoid generated by X .

Proposition 2.1.5. Let $X \subseteq M$. Then

$$\text{Mon}(X) = \{1_M x_1 x_2 \cdots x_n : n \in \mathbb{N} \cup \{0\}, x_i \in X\}.$$

Remark 2.1.19. Essentially, when we generate a submonoid of a monoid, we always include the identity of the monoid. If $X \subseteq M$ is such that $\text{Mon}(X) = M$, then X is a monoid generating set for M and M is generated by X as a monoid. Notice that if X is a generating set for M , then X is also a monoid generating set; on the other hand, if X is a monoid generating set for M , then $X \cup \{1_M\}$ is a generating set for M . Thus M is finitely generated if and only if there is a finite monoid generating set for M .

2.1.5 Ideals and Monogenic Semigroup

Definition 2.1.20 (Ideal). Let T be a subset of S . The subset T is a left ideal of S if it is closed under left multiplication by any element of S ; that is, if $ST \subseteq T$. It is a right ideal of S if it is closed under right multiplication by any element of S ; that is, if $TS \subseteq T$. It is a two-sided ideal, or simply an ideal, of S if it is closed under both left and right multiplication by elements of S ; that is, if $ST \cup TS \subseteq T$.

Every ideal, whether left, right, or two-sided, is a subsemigroup.

Definition 2.1.21 (Principal ideal). For any $x \in S$, define

$$L(x) = S^1 x = \{x\} \cup Sx, \quad R(x) = x S^1 = \{x\} \cup xS, \quad J(x) = S^1 x S^1 = \{x\} \cup xS \cup Sx \cup SxS.$$

Then $L(x)$, $R(x)$, and $J(x)$ are, respectively, the principal left ideal generated by x , the principal right ideal generated by x , and the principal ideal generated by x . As they are, respectively, a left ideal, a right ideal, and a (two-sided) ideal.

Example 2.1.22. 1. Consider the semigroup $(\mathbb{N}, +)$. Let $n \in \mathbb{N}$ and let

$$I_n = \{m \in \mathbb{N} : m \geq n\}.$$

Then I_n is an ideal of $\mathbb{N}$; indeed, $I_n = L(n) = R(n) = J(n)$.

2. Let S be a right zero semigroup. Let T be a subset of S . Then $ST = T$ since $xy = y$ for any $x \in S$ and $y \in T$. So T is a left ideal of S . On the other hand $TS = S$, and so T is a right ideal of S if and only if $T = S$.

3. Let G be a group. Let T be a subset of G . For any $x \in G$ and $y \in T$, we have $x = xy^{-1}y \in Gy$; hence $Gy = G$. So T is a left ideal if and only if $T = G$; similarly, T is a right ideal if and only if $T = G$. So the only left ideal and right ideal of G is G itself.

Definition 2.1.23 (Rees congruence and Rees quotient). Let S be a semigroup and I a proper ideal of S . The *Rees congruence* on S modulo I is

$$\rho_I = (I \times I) \cup 1_S,$$

so that $x \rho_I y$ if and only if $x = y$ or $x, y \in I$. The quotient semigroup S/ρ_I , also written as S/I and called the *Rees quotient* of S by I , may be identified with

$$S/I = \{I\} \cup \{\{x\} : x \in S \setminus I\},$$

which can be regarded as I together with the elements of $S \setminus I$.

In S/I the product of two elements in $S \setminus I$ is same as their product in S if this lies in $S \setminus I$; otherwise the product is I .

Since the element I is the zero element of the semigroup, another way of thinking S/ρ_I is as $(S \setminus I) \cup \{0\}$, where all products not falling in $S \setminus I$ are zero.

The map $\varphi : S \rightarrow S/I$ given by $x\varphi = x$ if $x \notin I$ and $x\varphi = 0$ if $x \in I$ is a surjective homomorphism, called the *natural homomorphism*.

Definition 2.1.24. Let $\varphi : S \rightarrow T$ be a homomorphism. Then, φ is said to be a *rees morphism* if $\ker \varphi$ is a rees congruence on S .

Definition 2.1.25 (Ideal extension). Let M be a semigroup and T a semigroup with zero 0 . A semigroup E is called an *ideal extension* of M by T if M is an ideal of E and the Rees quotient semigroup E/M is isomorphic to T .

Definition 2.1.26 (Monogenic semigroup). Suppose S is generated by a single element x ; that is, $S = \langle x \rangle$. Then by Proposition 2.1.4,

$$S = \{x^n : n \in \mathbb{N}\},$$

and we call S the monogenic semigroup generated by x .

There are two possibilities: either all the powers of x are distinct, or there exist $k, m \in \mathbb{N}$ such that $x^k = x^{k+m}$. In the latter case, x is said to be periodic. If k and m are minimal, then S is finite, with $k + m - 1$ elements,

$$x, x^2, \dots, x^{k-1}, x^k, x^{k+1}, \dots, x^{k+m-1}.$$

We call k the *index* of x and m the *period* of x .

Remark 2.1.27. Let $K = \{x^k, x^{k+1}, \dots, x^{k+m-1}\}$. It is easy to see that K is an ideal of S . Furthermore, K is a subgroup of S , since the integers $k, k+1, \dots, k+m-1$ are representatives for congruence classes of the integers modulo m . In particular, some power of x is the identity of this subgroup and is therefore an idempotent.

Definition 2.1.28. A periodic semigroup is one in which every element is periodic.

2.1.6 Homomorphisms

Definition 2.1.29. Let S and T be semigroups. A map $\varphi : S \rightarrow T$ is a *homomorphism* if

$$(xy)\varphi = (x\varphi)(y\varphi) \quad \forall \quad x, y \in S.$$

If S and T are monoids, then φ is a *monoid homomorphism* if

$$(xy)\varphi = (x\varphi)(y\varphi) \quad \forall \quad x, y \in S \quad \text{and} \quad 1_S\varphi = 1_T.$$

A *monomorphism* is an injective homomorphism. An *epimorphism* is a surjective homomorphism. If $\varphi : S \rightarrow T$ is an epimorphism, then T is a homomorphic image of S .

Definition 2.1.30. An *isomorphism* is a bijective homomorphism.

It is easy to prove that a homomorphism $\varphi : S \rightarrow T$ is an isomorphism if and only if there is a homomorphism $\varphi^{-1} : T \rightarrow S$ such that $\varphi\varphi^{-1} = id_T$ and $\varphi^{-1}\varphi = id_S$. If there is an isomorphism $\varphi : S \rightarrow T$, then we say that S and T are isomorphic, and denote this by $S \cong T$.

It is easy to prove that if $\varphi : S \rightarrow T$ is a homomorphism and S' and T' are subsemigroups of S and T , respectively, then $S'\varphi$ is a subsemigroup of T and $T'\varphi^{-1}$ is a subsemigroup of S . In particular, putting $S' = S$ shows that $\text{im } \varphi$ is a subsemigroup of T . If φ is a monomorphism, then S is isomorphic to the subsemigroup $\text{im } \varphi$ of T .

Definition 2.1.31. The *kernel* of a homomorphism $\varphi : S \rightarrow T$ is the binary relation

$$\ker \varphi = \{(x, y) \in S \times S : x\varphi = y\varphi\}.$$

Notice that φ is a monomorphism if and only if $\ker \varphi$ is the identity relation (that is, $\ker \varphi = id_S$).

The following result shows that every semigroup is isomorphic to a subsemigroup of a transformation semigroup. This is the semigroup analogue of Cayley's theorem for groups, which states that every group is isomorphic to a subgroup of a symmetric group.

Theorem 2.1.6. For any $x \in S$, let $\rho_x \in \mathcal{T}_{S^1}$ be the map defined by $s\rho_x = sx$ for all $s \in S^1$. Then the map $\varphi : S \rightarrow \mathcal{T}_{S^1}$ given by $x \mapsto \rho_x$ is a monomorphism of S .

Proof. Let $x, y, s \in S$. Then

$$s\rho_x\rho_y = (s\rho_x)\rho_y = (sx)\rho_y = (sx)y = s(xy) = s\rho_{xy}.$$

Hence $\rho_x\rho_y = \rho_{xy}$, and so $(xy)\varphi = (x\varphi)(y\varphi)$. Therefore φ is a homomorphism.

Furthermore, if $x\varphi = y\varphi$, then $\rho_x = \rho_y$. Hence

$$1\rho_x = 1\rho_y,$$

that is, $x = y$. Therefore φ is injective. □

A map $\varphi : S \rightarrow T$ is an *anti-homomorphism* if

$$(xy)\varphi = (y\varphi)(x\varphi)$$

for all $x, y \in S$.

Remark 2.1.32. An *endomorphism* is a homomorphism from a semigroup S to itself. The set of all endomorphisms on S is denoted $\text{End}(S)$ and forms a subsemigroup of $\mathcal{T}_S$.

A semigroup S is *group-embeddable* if there is a group G and a monomorphism $\varphi : S \rightarrow G$. In this case, S is isomorphic to the subsemigroup $\text{im } \varphi$ of G . Clearly any group-embeddable semigroup is cancellative, but we shall see that there exist cancellative semigroups that are not group-embeddable.

2.1.7 Subdirect products

Definition 2.1.33 (Projection maps). Let $\mathcal{S} = \{S_i : i \in I\}$ be a collection of semigroups. For each $j \in I$, there is a *projection map* from the direct product $\prod_{i \in I} S_i$ onto S_j , taking an element of the direct product to its j -th component:

$$\pi_j : \prod_{i \in I} S_i \rightarrow S_j, \quad x\pi_j = (j)x.$$

Notice that every π_j is an epimorphism.

Definition 2.1.34 (Subdirect product). A *subdirect product* of $\mathcal{S}$ is [a semigroup isomorphic to] a subsemigroup P of the direct product $\prod_{i \in I} S_i$ such that $P\pi_i = S_i$ for all $i \in I$.

Definition 2.1.35 (Separation by epimorphisms). Let S be a semigroup. A collection of epimorphisms $\Phi = \{\varphi_i : S \rightarrow S_i : i \in I\}$ is said to separate the elements of S if they have the property that

$$(\forall i \in I)(x\varphi_i = y\varphi_i) \implies x = y.$$

Proposition 2.1.7. A semigroup S is a subdirect product of a collection of semigroups $\mathcal{S} = \{S_i : i \in I\}$ if and only if there exists a collection of epimorphisms $\Phi = \{\varphi_i : S \rightarrow S_i : i \in I\}$ that separate the elements of S .

Proof. If S is a subdirect product of $\mathcal{S}$, then the collection of projection maps restricted to S (that is, $\{\pi_i|_S : S \rightarrow S_i : i \in I\}$) separates the elements of S .

On the other hand, suppose the collection Φ separates the elements of S . Define $\psi : S \rightarrow \prod_{i \in I} S_i$ by letting the i -th component of $s\psi$ be $s\varphi_i$; that is, $(i)(s\psi) = s\varphi_i$. Then ψ is a homomorphism since each φ_i is a homomorphism. Furthermore, $s\psi = t\psi$ implies $s\varphi_i = t\varphi_i$ for all $i \in I$, which means that Φ separates the elements of S . Hence ψ is injective, so S is isomorphic to the subsemigroup $\text{im } \psi$ of $\prod_{i \in I} S_i$. Finally, the projection maps π_i are all surjective since each φ_i is surjective. So $\text{im } \psi$ is a subdirect product of $\mathcal{S}$. $\square$

Proposition 2.1.8. Let S be a semigroup and let $\Sigma = \{\sigma_i : i \in I\}$ be a collection of congruences on S . Let $\sigma = \bigcap_{i \in I} \sigma_i$. Then S/σ is a subdirect product of $\{S/\sigma_i : i \in I\}$.

Proof. For each i , there is a homomorphism $\varphi_i : S/\sigma \rightarrow S/\sigma_i$ with $[x]_\sigma \varphi_i = [x]_{\sigma_i}$. These maps are well-defined by Lemma 1.1.13. Clearly, the homomorphisms φ_i are surjective. Furthermore, the collection $\Phi = \{\varphi_i : i \in I\}$ separates the elements of S/σ , since if $[x]_\sigma \varphi_i = [y]_\sigma \varphi_i$ for all $i \in I$, then $[x]_{\sigma_i} = [y]_{\sigma_i}$ for all $i \in I$, which implies $(x, y) \in \sigma$. Therefore S/σ is a subdirect product of $\{S/\sigma_i : i \in I\}$ by Proposition 2.1.7. $\square$

Proposition 2.1.9. Let M be a monoid and let E be an ideal extension of M by T . Then E is a subdirect product of M and T .

Proof. Let $\varphi : E \rightarrow T$ be the natural homomorphism $x\varphi = [x]_M$. Let $\psi : E \rightarrow M$ be given by $x\psi = x1_M$. Then

$$\begin{aligned} (x\psi)(y\psi) &= x1_M y1_M \\ &= xy1_M \\ &= (xy)\psi, \end{aligned}$$

since $y1_M$ lies in the ideal M of E . Thus ψ is a homomorphism. Both φ and ψ are clearly surjective. Furthermore, if $x\varphi = y\varphi$, then either $x, y \in E - M$ and $[x]_M = [y]_M$, so $x = y$, or $x, y \in M$ and $x1_M = y1_M$, so $x = y$. Thus the collection of epimorphisms $\{\varphi, \psi\}$ separates elements of E , and so E is a subdirect product of M and T . $\square$

2.1.8 Actions

Definition 2.1.36 (Semigroup action). A *semigroup action* of a semigroup S on a set A is an operation

$$\cdot : A \times S \rightarrow A$$

that is compatible with the semigroup multiplication, in the sense that

$$(a \cdot x) \cdot y = a \cdot (xy)$$

for all $a \in A$ and $x, y \in S$.

We call such a semigroup action an *action of S on A* , or an *S -action on A* , and say that S acts on A .

Example 2.1.37. 1. Any subsemigroup S of $\mathcal{T}(A)$ acts on A by $a \cdot \varphi = a\varphi$ (where $\varphi \in \mathcal{T}(A)$).

2. Let S be a subsemigroup of a semigroup T . Then S acts on T via $t \cdot x = tx$ for all $t \in T$ and $x \in S$. In particular, this holds when $T = S$ or when $T = S^1$.

Given an action, we can define a map $\varphi : S \rightarrow \mathcal{T}(A)$, where the transformation $s\varphi$ is such that $a(s\varphi) = a \cdot s$. The condition $(a \cdot x) \cdot y = a \cdot (xy)$ implies that φ is a homomorphism. Conversely, given a homomorphism $\varphi : S \rightarrow \mathcal{T}(A)$, we can define an action by $a \cdot s = a(s\varphi)$, which satisfies $(a \cdot x) \cdot y = a \cdot (xy)$. There is thus a one-to-one correspondence between actions of a semigroup S on A and homomorphisms $\varphi : S \rightarrow \mathcal{T}(A)$.

Definition 2.1.38 (Free action). An action of S on A is *free* if distinct elements of S act differently on every element of A , or, equivalently,

$$(\forall x, y \in S) \left((\exists a \in A) (a \cdot x = a \cdot y) \Rightarrow x = y \right).$$

Definition 2.1.39 (Transitive action). An action of S on A is *transitive* if for all $a, b \in A$, there exists some element $s \in S$ such that $a \cdot s = b$.

Definition 2.1.40 (Regular action). An action of S on A is *regular* if it is both free and transitive.

Suppose A is also a semigroup. An action of S on A is by endomorphisms if $s\varphi \in \text{End}(A)$ for each $s \in S$; in this case,

$$(ab) \cdot x = (a \cdot x)(b \cdot x)$$

for all $a, b \in A$ and $x \in S$.

Note 2.1.41. The above discussions concern a right semigroup action. There is a dual notion of a left semigroup action

Definition 2.1.42. A *left semigroup action* of S on A is an operation

$$\cdot : S \times A \rightarrow A$$

satisfying

$$s \cdot (t \cdot a) = (st) \cdot a.$$

This corresponds to a map $\varphi : S \rightarrow \mathcal{T}(A)$, where $a(s\varphi) = s \cdot a$. This map φ is an anti-homomorphism since

$$a(t\varphi)(s\varphi) = (t \cdot a)(s\varphi) = s \cdot (t \cdot a) = (st) \cdot a = a((st)\varphi).$$

The definitions of actions being free, transitive, regular, and by endomorphisms also apply to left actions.

Remark 2.1.43. The correspondence of right actions with homomorphisms and left actions with anti-homomorphisms depends on writing mappings on the right and composing them left to right. When mappings are written on the left and composed right to left, right actions correspond to anti-homomorphisms and left actions correspond to homomorphisms.

2.2 Free semigroups and presentations

Informally, the free semigroup on a set A is the unique most general semigroup generated by A , in the sense that every other semigroup generated by A is a homomorphic image (and hence a factor semigroup) of the free semigroup on A .

Definition 2.2.1. An alphabet A is an abstract set whose elements are called *letters*.

2.2.1 Free Semigroups

A word over A is a finite sequence

$$(a_1, a_2, \dots, a_n),$$

where $a_i \in A$. The length of this word is defined to be n .

The empty sequence is called the *empty word* and is denoted by ε ; its length is 0. We denote the set of all non-empty words over A by A^+ , and the set of all words, including the empty word, by A^* . That is

$$A^* = A^+ \cup \{\varepsilon\}.$$

The multiplication of words is defined by concatenation:

$$(a_1, \dots, a_n)(b_1, \dots, b_m) = (a_1, \dots, a_n, b_1, \dots, b_m).$$

Clearly, this multiplication is associative, and so A^+ forms a semigroup and A^* forms a monoid. The empty word ε is the identity of A^* .

Definition 2.2.2. The semigroup A^+ is called the *free semigroup* on A , and the monoid A^* as *free monoid* on A .

We shall usually write

$$a_1 a_2 \cdots a_n$$

instead of $(a_1, a_2, \dots, a_n)$, which is possible because of associativity. We denote the length of a word u by $|u|$. In particular,

$$|u| = 0 \iff u = \varepsilon.$$

Lemma 2.2.1. Let $u, v \in A^*$. Then

$$|uv| = |u| + |v|.$$

2.2.2 Results and Properties

Definition 2.2.3 (Equidivisible semigroup). A semigroup S is called *equidivisible* if, for all $x, y, z, t \in S$,

$$xy = zt$$

implies that either

$$(\exists p \in S)(x = zp \text{ and } t = py),$$

or

$$(\exists q \in S)(z = xq \text{ and } y = qt).$$

Every group is equidivisible. Indeed, if $xy = zt$, then one may take $p = z^{-1}x$ or $q = x^{-1}z$.

Proposition 2.2.2. *The free semigroup A^* is equidivisible.*

Proof. Suppose $x, y, z, t \in A^*$ are such that $xy = zt$. Let

$$xy = zt = a_1 \cdots a_n,$$

where $a_i \in A$. Then, by the definition of multiplication in A^* , we have

$$x = a_1 \cdots a_k, \quad y = a_{k+1} \cdots a_n, \quad z = a_1 \cdots a_\ell, \quad t = a_{\ell+1} \cdots a_n,$$

where $0 \leq k, \ell \leq n+1$. We allow the values 0 and $n+1$ and formally take subwords $a_i \cdots a_j$, where $j < i$, to mean the empty word ε .

If $k \leq \ell$, let

$$q = a_{k+1} \cdots a_\ell.$$

Then $z = xq$ and $y = qt$.

On the other hand, if $k \geq \ell$, let

$$p = a_{\ell+1} \cdots a_k.$$

Then $x = zp$ and $t = py$. □

Proposition 2.2.3. *Let $A = \{x, y\}$. Let $B = \{b_i : i \in \mathbb{N}\}$. Then A^* contains a submonoid isomorphic to B^* .*

Proof. Define a map $\varphi : B \rightarrow A^*$ by

$$b_i \varphi = xy^i x.$$

Since B^* is free on B , this map φ extends to a unique homomorphism, which we also denote by φ , from B^* to A^* .

Suppose, with the aim of obtaining a contradiction, that φ is not injective. Then there exist $u, v \in B^*$ with

$$u\varphi = v\varphi.$$

Suppose u and v begin with the same symbol b_i ; that is, $u = b_i u'$ and $v = b_i v'$. Then

$$(b_i \varphi)(u' \varphi) = (b_i \varphi)(v' \varphi)$$

and so $u' \varphi = v' \varphi$ by cancellativity in A^* . Thus we can replace u by u' and v by v' and repeat this process until we have words u and v beginning with different symbols. Therefore assume that u and v begin with symbols b_i and b_j , respectively, where $i \neq j$; that is,

$$u = b_i u', \quad v = b_j v'.$$

Then

$$xy^i x(u' \varphi) = (b_i \varphi)(u' \varphi) = (b_j \varphi)(v' \varphi) = xy^j x(v' \varphi).$$

Assume $i > j$; the other case is similar. By cancellativity in A^* , we have

$$y^{i-j} x(u' \varphi) = x(v' \varphi),$$

which is a contradiction since $i - j > 0$. Therefore φ is injective and so B^* is isomorphic to $\text{im } \varphi$. $\square$

A similar result holds for free semigroups.

Definition 2.2.4 (Indecomposable element). An element a of a semigroup S is said to be *indecomposable* if there do not exist $u, v \in S$ such that

$$a = uv.$$

Proposition 2.2.4. Let M be a submonoid of A^* , and let

$$N = M \setminus \{\varepsilon\}.$$

Then $N \setminus N^2$ is the unique minimal generating set for M .

Proof. Clearly, any generating set for M must contain $N - N^2$. So we must show that

$$\langle N - N^2 \rangle = M.$$

Clearly,

$$\langle N - N^2 \rangle \subseteq M;$$

we have to prove that

$$M \subseteq \langle N - N^2 \rangle.$$

We already know that $\varepsilon \in \langle N - N^2 \rangle$, so it remains to show that

$$N \subseteq \langle N - N^2 \rangle.$$

We proceed by induction on the length of elements of N . Let

$$k = \min\{|u| : u \in N\}.$$

Then if $|u| = k$ and $u \in N$, then by Lemma 2.2.1, u cannot be factored as a product of two elements of N . So

$$u \in N - N^2 \subseteq \langle N - N^2 \rangle.$$

Thus all words of length k in N lie in $\langle N - N^2 \rangle$.

So assume that all words of length less than l in N lie in $\langle N - N^2 \rangle$. Let $u \in N$ with $|u| = l$. If $u \in N - N^2$, then

$$u \in \langle N - N^2 \rangle.$$

On the other hand, if $u \notin N - N^2$, then $u \in N^2$ and so

$$u = u' u''$$

for $u', u'' \in N$. By Lemma 2.2.1,

$$|u'|, |u''| < |u| = l.$$

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So, by assumption,

$$u', u'' \in \langle N - N^2 \rangle$$

and hence

$$u \in \langle N - N^2 \rangle.$$

Therefore, by induction,

$$N \subseteq \langle N - N^2 \rangle.$$

Hence

$$M = \langle N - N^2 \rangle.$$

Therefore $N - N^2$ is a generating set for M , and since every generating set must contain $N - N^2$, it is the unique minimal generating set. $\square$

The elements of $N \setminus N^2$ are precisely the *indecomposable* elements of M .

Definition 2.2.5. The base of a submonoid or a subsemigroup M of A^* is defined to be $N - N^2$, where $N = M - \{\varepsilon\}$.

The base is the unique minimal generating set for M .

Corollary 2.2.5. The alphabet A is the base of both A^+ and A^* .

Proposition 2.2.6. Let $u, v \in A^*$. Then

$$uv = vu$$

if and only if there exist $w \in A^*$ and $i, j \in \mathbb{N}$ such that

$$u = w^i \quad \text{and} \quad v = w^j.$$

Proof. If $u = w^i$ and $v = w^j$, then

$$uv = w^{i+j} = vu.$$

To prove the other direction, we proceed by induction on $|uv|$. If

$$|uv| \leq 2,$$

then clearly $uv = vu$ implies $u = v$. So assume the result holds for $|uv| < k$. Suppose $uv = vu$. Interchanging u and v if necessary, assume

$$|u| \geq |v|.$$

By Proposition 2.2.2, there exists $p \in A^*$ such that

$$u = vp \quad \text{and} \quad u = pv.$$

If $p = \varepsilon$, then $u = v$. Otherwise,

$$vp = pv,$$

where

$$|vp| < |uv|.$$

So by induction,

$$v = w^j \quad \text{and} \quad p = w^i$$

for some $w \in A^*$ and $i, j \in \mathbb{N}$. Thus

$$u = vp = w^{i+j}.$$

Hence, by induction, the result holds for all $u, v \in A^+$. $\square$

Proposition 2.2.7. *A semigroup S is free if and only if every element of S has a unique representative as a product of elements of $S - S^2$.*

Proof. Clearly every element of A^+ has a unique representative as a product of elements of A . So suppose every element of S has a unique representative as a product of elements of $A = S - S^2$.

We will show that S satisfies the definition of freedom. Let T be a semigroup and $\varphi : A \rightarrow T$ a map. Define a map $\varphi^+ : S \rightarrow T$ by letting

$$s\varphi^+ = (a_1\varphi)(a_2\varphi) \cdots (a_n\varphi),$$

where $a_1a_2 \cdots a_n$ is the unique representative of s as a product of elements $a_i \in A$.

Notice that if $t \in S$ is uniquely represented by $b_1 \cdots b_m$, where $b_i \in A$, then st has unique representative

$$a_1 \cdots a_nb_1 \cdots b_m.$$

Hence φ^+ is a homomorphism. It is clear that φ^+ is the unique homomorphism extending φ , and so S is free on A . $\square$

Example 2.2.6. Let $A = \{x\}$ and let $S = \langle x^2, x^3 \rangle$.

Then

$$S - S^2 = \{x^2, x^3\}.$$

But $x^5 \in S$ and

$$x^5 = x^2x^3 = x^3x^2,$$

so x^5 has two distinct representatives as a product of elements of $S - S^2$. Hence S is not free.

The above example shows that a free semigroup contains subsemigroups which are not free. In contrast, every subgroup of a free group is free by the Nielsen-Schreier theorem.

Definition 2.2.7 (Length function). A *length function* for a semigroup S is a homomorphism $s \mapsto |s|$. A proper length function for a monoid M is a length function with the additional property that

$$|s| = 0 \implies s = 1_M.$$

Notice that the usual notion of length for free semigroups as defined previously is also a proper length function.

Proposition 2.2.8. *A monoid M is equidivisible and has a proper length function if and only if it is isomorphic to A^* .*

Proof. We have already seen that A^* is equidivisible and has a proper length function. So suppose that M is an equidivisible monoid with a proper length function. Let

$$N = M - \{1_M\}.$$

Notice that N is a subsemigroup since

$$x, y \in N \implies (|x| \neq 0 \wedge |y| \neq 0) \implies |xy| = |x| + |y| \neq 0 \implies xy \neq 1_M.$$

Let $A = N - N^2$ and $w \in N^2$. Then $w = w'w''$ with

$$0 < |w'| < |w| \quad \text{and} \quad 0 < |w''| < |w|.$$

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Continuing this factoring process, we get

$$w = a_1 \cdots a_n,$$

where $a_i \in A$. So every element of M has a representative as a product of elements of A . Suppose w has another such representative $b_1 \cdots b_m$. We aim to show by induction on $m + n$ that the two representatives are identical. If $m + n = 2$, then

$$a_1 = w = b_1.$$

So suppose the result holds for all $m + n < k$.

Since

$$a_1 \cdots a_n = b_1 \cdots b_m,$$

equidivisibility shows that there exists $p \in M$ with either

$$a_1 = b_1 p \quad \text{and} \quad p a_2 \cdots a_n = b_2 \cdots b_m,$$

or

$$b_1 = a_1 p \quad \text{and} \quad p b_2 \cdots b_m = a_2 \cdots a_n.$$

Since $a_1 \in A = N - N^2$, it cannot be written as a product of two non-identity elements. Thus in the first case $p = 1_M$, and so

$$a_1 = b_1, \quad a_2 \cdots a_n = b_2 \cdots b_m.$$

The second case is similar. Applying the induction hypothesis shows

$$m = n$$

and

$$a_i = b_i$$

for all $i \geq 2$.

Thus every element of N has a unique representative as a product of elements of A , and so $N \cong A^+$ by Proposition 2.2.7. Hence

$$M = N \cup \{1_M\} \cong A^*.$$

□

Definition 2.2.8. Let x be an element of a monoid M . A non-trivial left factor of x is an element y such that $x = yz$ for some $z \in M - \{1_M\}$.

Proposition 2.2.9. A monoid M is free if it has the following four properties:

1. the only left- and right-invertible element of M is 1_M ;
2. M is cancellative;
3. M is equidivisible;
4. no element of M has infinitely many distinct non-trivial left factors.

Proof. Clearly, A^* has the four properties. So suppose M has the four properties (1)–(4). We first show that $N = M - \{1_M\}$ is a subsemigroup of M .

Let $x, y \in M$ with $xy = 1_M$. Then

$$(yx)^2 = y(xy)x = yx.$$

So yx is an idempotent. Since M is cancellative, $yx = 1_M$. Thus x and y are left and right inverses of each other, and hence

$$x = y = 1_M.$$

Therefore $x, y \in N$ implies $xy \in N$, and so N is a subsemigroup.

Now suppose

$$N = N^2.$$

Pick $x \in N$. Then there exists some m such that x has exactly m distinct non-trivial left factors. But since

$$N^{m+2} = N,$$

we can write

$$x = y_1 \cdots y_{m+2}$$

for $y_i \in N$. The products

$$y_1 \cdots y_k, \quad k < m + 2,$$

are all non-trivial left factors of x . Furthermore, they are all distinct, for if

$$y_1 \cdots y_k = y_1 \cdots y_{k+1},$$

then cancellativity shows that

$$y_{k+1} \cdots y_{m+2} = 1_M,$$

which is impossible since N is a subsemigroup. Thus x has at least $m + 1$ distinct non-trivial left factors, which is a contradiction. Hence $N^2 \subsetneq N$.

Thus $N - N^2$ is non-empty.

Similarly, suppose

$$x \in \bigcap \{N^m : m \in \mathbb{N}\}.$$

Suppose x has exactly m distinct non-trivial left factors. Since $x \in N^{m+2}$, the same contradiction arises. So $\bigcap \{N^m : m \in \mathbb{N}\}$ is empty. Therefore

$$N \supsetneq N^2 \supsetneq N^3 \supsetneq \cdots,$$

and hence every element $x \in N$ lies in $N^m - N^{m+1}$ for some unique m , and so x has an expression

$$x = n_1 \cdots n_m$$

as a product of elements $n_i \in N$.

Arguing by equidivisibility as in the proof of Proposition 2.2.8, we see that each element has a unique expression of this form. So M is free by Proposition 2.2.7. $\square$

2.2.3 Universal Property

Let F be a semigroup and let A be an alphabet. Let $\iota : A \hookrightarrow F$ be an embedding of A into F . Then the semigroup F is *free on A* if, for any semigroup S and map $\varphi : A \rightarrow S$, there is a unique homomorphism $\varphi^+ : F \rightarrow S$ extending φ . That is,

$$\iota\varphi^+ = \varphi,$$

or, equivalently, the following diagram commutes:

$$\begin{array}{ccc} A & \xrightarrow{\iota} & F \\ & \searrow \varphi & \downarrow \varphi^+ \\ & & S \end{array}$$

Proposition 2.2.10 (Uniqueness of free semigroups). *Let A be an alphabet and let F be a semigroup. Then F is free on A if and only if $F \cong A^+$.*

Proof. Suppose $F \cong A^+$. To show F is free on A , it is sufficient to show that A^+ is free on A . Let

$$\iota : A \rightarrow A^+$$

be the embedding map. Let S be a semigroup and $\varphi : A \rightarrow S$ be a map. Define

$$\varphi^+ : A^+ \rightarrow S$$

by

$$(a_1 a_2 \cdots a_n)\varphi^+ = (a_1\varphi)(a_2\varphi) \cdots (a_n\varphi).$$

It is easy to see that φ^+ is a homomorphism and that $\iota\varphi^+ = \varphi$.

We now have to prove that φ^+ is unique. So let

$$\psi : A^+ \rightarrow S$$

be an arbitrary homomorphism with $\iota\psi = \varphi$.

For any $a_1 a_2 \cdots a_n \in A^+$,

$$\begin{aligned} (a_1 a_2 \cdots a_n)\psi &= (a_1\psi)(a_2\psi) \cdots (a_n\psi) && \text{[since } \psi \text{ is a homomorphism]} \\ &= (a_1\varphi^+)(a_2\varphi^+) \cdots (a_n\varphi^+) && \text{[since } \iota\psi = \varphi = \iota\varphi^+ \text{]} \\ &= (a_1 a_2 \cdots a_n)\varphi^+ && \text{[since } \varphi^+ \text{ is a homomorphism].} \end{aligned}$$

Thus $\psi = \varphi^+$. Hence φ^+ is the unique homomorphism from A^+ to S with $\iota\varphi^+ = \varphi$, and so A^+ is free on A .

Now let F be a semigroup that is free on A . Let

$$\iota_1 : A \rightarrow A^+ \quad \text{and} \quad \iota_2 : A \rightarrow F$$

be the embedding maps. Put $\varphi = \iota_2$ and $S = F$ in the definition of F being free on A to see that there is a homomorphism $\iota_2^+ : A^+ \rightarrow F$ with $\iota_1\iota_2^+ = \iota_2$.

Similarly, since F is free on A , there is a homomorphism $\iota_1^+ : F \rightarrow A^+$ with $\iota_2\iota_1^+ = \iota_1$. Therefore

$$\iota_1 = \iota_2\iota_1^+ \quad \text{and} \quad \iota_2 = \iota_1\iota_2^+.$$

That is, the following diagrams commute:

$$\begin{array}{ccc} A & \xrightarrow{\iota_1} & A^+ \\ & \searrow \iota_2 & \downarrow \iota_2^+ \\ & & F \end{array} \quad \begin{array}{ccc} A & \xrightarrow{\iota_2} & F \\ & \searrow \iota_1 & \downarrow \iota_1^+ \\ & & A^+ \end{array}$$

Therefore, by the uniqueness requirement in the definition of freedom,

$$\iota_2^+ \iota_1^+ = id_{A^+} \quad \text{and} \quad \iota_1^+ \iota_2^+ = id_F.$$

Hence ι_1^+ and ι_2^+ are mutually inverse isomorphisms, and so

$$F \cong A^+.$$

□

Remark 2.2.9. What makes free semigroups interesting is that every semigroup is isomorphic to a quotient of a free semigroup.

To see this, let $\varphi : A \rightarrow S$ be such that $\text{im } \varphi$ generates S (e.g., we could choose A to have the same cardinality as S and φ to be a bijection). Then φ^+ extends to a homomorphism $\varphi^+ : A^+ \rightarrow S$. Since $\langle \text{im } \varphi \rangle = S$, we have $\text{im } \varphi^+ = S$.

By the Homomorphism Theorem 1.1.7,

$$\begin{array}{ccc} A^+ & \xrightarrow{\varphi^+} & S \\ \downarrow \pi_{\ker \varphi^+} & \nearrow \varphi^+ & \\ A^+ / \ker \varphi^+ & & \end{array} \quad \text{and hence}$$

$$A^+ / \ker \varphi^+ \cong \text{im } \varphi^+ = S.$$

Thus, as expected, S is isomorphic to a quotient of a free semigroup.

Instead of choosing a semigroup S and knowing that it is a quotient of a free semigroup, we can start with a free semigroup and impose a congruence σ on it to obtain the quotient semigroup A^+ / σ . This is the idea of semigroup presentations, which we shall discuss next, but first we have a lemma which will be useful later in the discussion of presentations.

Lemma 2.2.11. For any $\rho \in \mathfrak{B}(S)$, we have

$$\rho^c = \{(pxq, pyq) \in S \times S : p, q \in S^1, (x, y) \in \rho\}.$$

Proof. Let

$$\sigma = \{(pxq, pyq) \in S \times S : p, q \in S^1, (x, y) \in \rho\}.$$

To prove $\sigma = \rho^c$, we have to show that σ is the smallest left and right compatible relation on S containing ρ .

Notice first that if $(x, y) \in \rho$, then

$$(x, y) = (1x1, 1y1) \in \sigma.$$

Hence σ contains ρ .

Let $(u, v) \in \sigma$ and $r \in S$. Then $u = pxq$ and $v = pyq$ for some $(x, y) \in \rho$. Let $p' = rp$. Then

$$(ru, rv) = (p'xq, p'yq) \in \sigma.$$

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Hence σ is left compatible and similarly right compatible.

Now let τ be some left and right compatible relation that contains ρ . Let $(pxq, pyq) \in \sigma$, where $(x, y) \in \rho$ and $p, q \in S^1$. Then $(x, y) \in \tau$ since $\rho \subseteq \tau$. Hence

$$(pxq, pyq) \in \tau$$

since τ is left and right compatible. Hence $\sigma \subseteq \tau$.

Hence σ is the smallest left and right compatible relation containing ρ . Therefore $\sigma = \rho^c$. $\square$

2.2.4 Presentations

Now we introduce the idea of a semigroup presentation which, as mentioned previously, is to define a semigroup as a quotient of a free semigroup, that is, as A^+/σ for some congruence σ on the free semigroup A^+ .

By Proposition 2.2.10, in order to specify the free semigroup it is sufficient to specify the alphabet A . In order to specify the congruence σ , it is sufficient to specify some binary relation ρ which generates σ .

Definition 2.2.10 (Semigroup presentation). A *semigroup presentation* is a pair $\langle A \mid \rho \rangle$, where A is an alphabet and ρ is a binary relation on A^+ . This presentation defines, or presents, any semigroup isomorphic to $A^+/\rho^\#$, where $\rho^\#$ is the congruence generated by ρ .

If both A and ρ are finite, then the presentation is said to be *finite*, and the semigroup presented by it is said to be *finitely presented*.

The elements of ρ are called *defining relations*. We think of the semigroup presented by $\langle A \mid \rho \rangle$ as the largest semigroup generated by A and satisfying the defining relations in ρ .

If $S = \langle A \mid \rho \rangle$, we view words in A^+ as representing elements of S . In particular, A represents a generating set for S .

In general, an element of S may have different representatives in A^+ . If $u, v \in A^+$ represent the same element of S , that is, if $(u, v) \in \rho^\#$, we write

$$u =_S v.$$

Definition 2.2.11. An elementary ρ -transition is a pair $(u, v) \in (\rho^S)^c$. Thus, by Lemma 2.2.11, (u, v) is an elementary ρ -transition if and only if v can be obtained from u by substituting a subword y for a subword x of u , where $(x, y) \in \rho$ or $(y, x) \in \rho$.

Proposition 2.2.12. Let S be presented by $\langle A \mid \rho \rangle$. Then $u =_S v$ if and only if there is a sequence

$$u = u_0, u_1, \dots, u_m = v$$

in A^+ such that each pair (u_i, u_{i+1}) is an elementary ρ -transition.

Proof. First of all, notice that

$$\begin{aligned}
 u &=_{\mathcal{S}} v \\
 \iff (u, v) &\in \rho^{\#} \\
 \iff (u, v) &\in (\rho^c)^e \\
 \iff (u, v) &\in (((\rho^c)^R)^S)^{\infty} \\
 \iff (u, v) &\in (\rho^c \cup (\rho^c)^{-1} \cup id_{A^+})^{\infty} \\
 \iff (u, v) &\in \bigcup_{n=1}^{\infty} (\rho^c \cup (\rho^c)^{-1} \cup id_{A^+})^n \\
 \iff (\exists n \in \mathbb{N}) &((u, v) \in (\rho^c \cup (\rho^c)^{-1} \cup id_{A^+})^n) \\
 \iff (\exists n \in \mathbb{N}) \text{ and } &(u = u_0, \dots, u_n = v) ((u_i, u_{i+1}) \in (\rho^c \cup (\rho^c)^{-1} \cup id_{A^+})).
 \end{aligned}$$

Hence $u =_{\mathcal{S}} v$ if and only if there is a sequence $u_0, \dots, u_n$ with $u_0 = u$, $u_n = v$, and each (u_i, u_{i+1}) being an *elementary ρ -transition* or with $u_i = u_{i+1}$.

Since we can shorten the sequence by removing any steps with $u_i = u_{i+1}$, we get that $u =_{\mathcal{S}} v$ if and only if there is such a sequence consisting entirely of *elementary ρ -transitions*. $\square$

When there is a sequence of elementary ρ -transitions from u to v , we say that (u, v) is a *consequence* of ρ , or that $u =_{\mathcal{S}} v$ can be deduced from the defining relations.

Definition 2.2.12 (Monoid presentation). The analogous discussion applies to monoids. A monoid M is free on A if and only if $M \cong A^*$. As for semigroups, every monoid is a quotient of a free monoid.

A monoid presentation $\langle A \mid \rho \rangle$ consists of an alphabet A and a binary relation ρ on A^* , and presents a monoid isomorphic to $A^*/\rho^{\#}$.

In a monoid presentation $\langle A \mid \rho \rangle$, the defining relations may involve the empty word ε . Thus relations in ρ may be of the form (u, ε) or (ε, u) .

If $M = \langle A \mid \rho \rangle$, then, for $u, v \in A^*$, $u =_M v$ if and only if there is a sequence of elementary ρ -transitions from u to v .

Proposition 2.2.13. *Suppose S is finitely presented and that $\langle A \mid \rho \rangle$ is a presentation for S . Then there exist $B \subseteq A$ and $\sigma \subseteq \rho$ such that $\langle B \mid \sigma \rangle$ is a finite presentation for S .*

Remark 2.2.13. The presentation $\langle A \mid \emptyset \rangle$, where there are no defining relations, defines the free semigroup A^+ .

Indeed, there are no non-trivial consequences of defining relations, and so all words over A represent different elements of the semigroup by Proposition 2.2.10. Alternatively, we may note that $\emptyset^{\#} = id_{A^+}$ and hence $A^+/id_{A^+} \cong A^+$.

Example 2.2.14. (a) The presentation $\langle a \mid \{(a^2, a)\} \rangle$ defines a trivial semigroup, since $a^n = a$ is the consequence of the defining relation.

(b) The presentation $\langle A \mid \{(ab, a) : a, b \in A\} \rangle$ defines a left zero semigroup on the set A .

(c) Let M be a monoid presented by $\langle a, b \mid (ab, ba) \rangle$. Then every element of M is of the form $a^i b^j$. So $M \cong (\mathbb{N} \cup \{0\}) \times (\mathbb{N} \cup \{0\})$.

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Definition 2.2.15 (Bicyclic monoid). The monoid presentation $\langle b, c \mid (bc, \varepsilon) \rangle$ defines the *bicyclic monoid*. Every element of the bicyclic monoid is represented by a word $c^\alpha b^\beta$, where $\alpha, \beta \in \mathbb{N} \cup \{0\}$, because if we have a word $w \in \{b, c\}^*$ that contains a subword bc , we can replace bc by ε . This process must end with a word that does not contain bc ; that is, a word $c^\alpha b^\beta$. In fact, each element is represented by a unique word of this form.

Example 2.2.16. Let $X = \{xyz, yz, yt, xy, zy, zt\} \subseteq \{x, y, z, t\}^+$ and let S be the subsemigroup of $\{x, y, z, t\}^+$ generated by X .

Let $A = \{a, b, c, d, e, f\}$ and $\varphi : A^+ \rightarrow S$ such that

$$\begin{aligned} a\varphi &= xyz, b\varphi = yz, c\varphi = yt, \\ d\varphi &= xy, e\varphi = zy, \text{ and } f\varphi = zt. \end{aligned}$$

Now, for any $\alpha \in \mathbb{N} \cup \{0\}$,

$$(ab^\alpha c)\varphi = (a\varphi)(b\varphi)^\alpha(c\varphi) = (xyz)(yz)^\alpha(yt) = x(yz)^{\alpha+1}yt = xy(zy)^\alpha t = (de^\alpha f)\varphi.$$

Hence, if $\langle A \mid \rho \rangle$ is a presentation for S , then $\rho^\#$ must contain the relation $(ab^\alpha c, de^\alpha f)$, for every $\alpha \in \mathbb{N} \cup \{0\}$.

Then there must be an elementary sequence of ρ -transitions from $ab^\alpha c$ to $de^\alpha f$. But ab^α is the unique word over A representing $(ab^\alpha)\varphi = x(yz)^{\alpha+1}$, and αc is the unique word representing $(\alpha c)\varphi = y(zy)^\alpha t$.

Hence, in an elementary sequence of ρ -transitions from $ab^\alpha c$ to $de^\alpha f$, the first must involve a defining relation from ρ , whose one side is $ab^\alpha c$. Thus ρ must contain infinitely many defining relations. Hence S is finitely generated, but not finitely presented.

The above example shows that a semigroup might be finitely generated but not finitely presented.

Example 2.2.17. Let $A = \{a, b, c, d, e, f, g, h\}$ and let $\rho = \{(ab, ba), (cf, de), (dg, ch)\}$.

Let S be the semigroup presented by $\langle A \mid \rho \rangle$.

Now suppose that $cu =_S cv$. Then there is an elementary ρ -transition sequence $cu = w_0, w_1, \dots, w_n = cv$.

Without loss of generality, we may assume that n be minimal among all such sequences.

Assume that c is altered at some step in this sequence. Then this step must involve applying one of the defining relations (cf, de) or (dg, ch) . Assume the former (latter is similar), then for some i , $w_i = cfw'$ and $w_{i+1} = dew'$.

Now, no defining relation begins with the symbol e , then e must occur in the terms of the sequence until the relation (cf, de) is applied again (as the symbol d must be removed because $w_n = cv$). Because the symbols de are not involved in any of the intermediate steps, there is no need to apply the (cf, de) twice. That is, there is a shorter sequence of elementary ρ -transitions from cu to cv , a contradiction to the minimality of n . Hence the initial symbol c is never altered. Thus we can delete the initial symbol c from each step in the sequence, and so $u \equiv_S v$.

Reasoning in this way for every symbol in A shows that S is left cancellative. Similarly, S is right cancellative. Thus S is cancellative.

Suppose S is group embeddable. Then there is a monomorphism $\varphi : S \rightarrow G$, where G is a

group. Then

$$\begin{aligned}
(ag)\varphi &= (a\varphi)(g\varphi) \\
&= (a\varphi)(e\varphi)(e\varphi)^{-1}(g\varphi) \\
&= (b\varphi)(f\varphi)(e\varphi)^{-1}(g\varphi) \quad (asae =_S bf) \\
&= (b\varphi)(c\varphi)^{-1}(c\varphi)(f\varphi)(e\varphi)^{-1}(g\varphi) \\
&= (b\varphi)(c\varphi)^{-1}(d\varphi)(e\varphi)(e\varphi)^{-1}(g\varphi) \quad (ascf =_S de) \\
&= (b\varphi)(c\varphi)^{-1}(d\varphi)(g\varphi) \\
&= (b\varphi)(c\varphi)^{-1}(c\varphi)(h\varphi) \quad (asdg =_S ch) \\
&= (b\varphi)(h\varphi) \\
&= (bh)\varphi
\end{aligned}$$

But $ag \neq bh$, there is no sequence of ρ -transitions from ag to bh since ag does not contain a subword that forms one side of defining relation in ρ . A contradiction arises, and so no such φ exists. Hence S is not group embeddable.

This example shows that cancellativity is not sufficient condition for group embeddability.

3 Green's Relations and Basic Structure

Green's relations, first studied by J. A. Green, are one of the fundamental concepts in semigroup theory. They help us understand the structure of a semigroup.

3.1 Green's equivalences

As already introduced, we denote the semigroup S^1 by $S \cup \{1\}$ if S has no identity, and S otherwise. For any $a \in S$, aS^1 is the right ideal generated by a , and S^1a is the left ideal generated by a . The smallest left ideal of S containing a is $S^1a = Sa \cup \{a\}$, and similarly for the right ideal.

We define an equivalence $\mathcal{L}$ on S by the rule that $a\mathcal{L}b$ if and only if $S^1a = S^1b$, that is, if a and b generate the same left ideal. Similarly, we define $\mathcal{R}$ by the rule that $a\mathcal{R}b$ if and only if $aS^1 = bS^1$.

Proposition 3.1.1. *Let S be a semigroup and $a, b \in S$. Then $a\mathcal{L}b$ if and only if there exist $x, y \in S^1$ such that $xa = b$ and $yb = a$. Also, $a\mathcal{R}b$ if and only if there exist $u, v \in S^1$ such that $au = b$ and $bv = a$.*

Another immediate property of $\mathcal{L}$ and $\mathcal{R}$ is:

Proposition 3.1.2. *$\mathcal{L}$ is a right congruence and $\mathcal{R}$ is a left congruence.*

We denote the intersection of $\mathcal{L}$ and $\mathcal{R}$ by $\mathcal{H}$. That is, $\mathcal{H} = \mathcal{L} \cap \mathcal{R}$, which is again an equivalence. The smallest equivalence relation which contains both $\mathcal{L}$ and $\mathcal{R}$ is denoted by $\mathcal{D}$. That is,

$$\mathcal{D} = \mathcal{L} \vee \mathcal{R}.$$

To make the relation easier to use, we have the following lemma.

Lemma 3.1.3. *The relations $\mathcal{L}$ and $\mathcal{R}$ commutes; that is,*

$$\mathcal{L} \circ \mathcal{R} = \mathcal{R} \circ \mathcal{L}.$$

Proof. Let S be a semigroup, let $a, b \in S$, and suppose that $(a, b) \in \mathcal{L} \circ \mathcal{R}$. Then there exists $c \in S$ such that $a\mathcal{L}c$ and $c\mathcal{R}b$. That is, there exist $x, y, u, v \in S^1$ such that

$$\begin{aligned} xa &= c, & cu &= b, \\ yc &= a, & bv &= c. \end{aligned}$$

If we now write d for the element ycu of S , we see that

$$au = ycu = d, \quad dv = ycuv = ybv = yc = a;$$

hence $a\mathcal{R}d$. Also,

$$yb = ycu = d, \quad xd = xycu = xau = cu = b,$$

and so $d\mathcal{L}b$. We deduce that $(a, b) \in \mathcal{R} \circ \mathcal{L}$.

We have shown that $\mathcal{L} \circ \mathcal{R} \subseteq \mathcal{R} \circ \mathcal{L}$; the reverse inclusion follows in a similar way. $\square$

Then, it follows from Proposition 1.1.12 that

$$\mathcal{D} = \mathcal{L} \circ \mathcal{R} = \mathcal{R} \circ \mathcal{L} = \mathcal{L} \vee \mathcal{R}.$$

Finally, the relation $\mathcal{J}$ is defined by $a\mathcal{J}b$ if and only if

$$S^1aS^1 = S^1bS^1,$$

that is, if a and b generate the same principal two-sided ideal.

As for $\mathcal{L}$ and $\mathcal{R}$, we have the following immediate property of $\mathcal{J}$:

Proposition 3.1.4. *Let S be a semigroup and $a, b \in S$. Then $a\mathcal{J}b$ if and only if there exist $x, y, u, v \in S^1$ such that $axy = b$ and $ubv = a$.*

Notice that $\mathcal{L} \subseteq \mathcal{J}$ and $\mathcal{R} \subseteq \mathcal{J}$. Since $\mathcal{D}$ is the smallest equivalence containing both $\mathcal{L}$ and $\mathcal{R}$, $\mathcal{D} \subseteq \mathcal{J}$.

On a group G , we have

$$\mathcal{H} = \mathcal{L} = \mathcal{R} = \mathcal{D} = \mathcal{J} = G \times G.$$

And on a commutative semigroup,

$$\mathcal{H} = \mathcal{L} = \mathcal{R} = \mathcal{D} = \mathcal{J}.$$

Proposition 3.1.5. *In a periodic semigroup, $\mathcal{J}$ and $\mathcal{D}$ coincide.*

Proof. We only need to show one side of inclusion.

Suppose that $a, b \in S$ are such that $a\mathcal{J}b$. Then there exist $x, y, u, v \in S^1$ such that $xay = b$ and $ubv = a$.

To prove the desired result we need to find an element c in S such that $a\mathcal{L}c$, $c\mathcal{R}b$. It follows easily from the equations above that

$$a = (ux)a(yv) = (ux)^2a(yv)^2 = (ux)^3a(yv)^3 = \dots$$

$$b = (xu)b(vy) = (xu)^2b(vy)^2 = (xu)^3b(vy)^3 = \dots$$

Since S is periodic we can find an m for which $(ux)^m$ is idempotent. Then, if we let $c = xa$, we find that

$$a = (ux)^ma(yv)^m = (ux)^m(ux)^ma(yv)^m = (ux)^ma = (ux)^{m-1}uxa = (ux)^{m-1}uc,$$

and so $a\mathcal{L}c$. Also, $cy = xay = b$, and if we choose n so that $(vy)^n$ is idempotent, we have

$$\begin{aligned} c &= xa = x(ux)^{n+1}a(yv)^{n+1} = (xu)^{n+1}xay(vy)^nv \\ &= (xu)^{n+1}b(vy)^{2n}v = (xu)^{n+1}b(vy)^{n+1}(vy)^{n-1}v \\ &= b(vy)^{n-1}v. \end{aligned}$$

Hence $c\mathcal{R}b$ as required. $\square$

The $\mathcal{L}$ -class, $\mathcal{R}$ -class, $\mathcal{D}$ -class, and $\mathcal{J}$ -class containing the element a will be denoted by L_a , R_a , D_a , and J_a , respectively.

Since $\mathcal{L}$, $\mathcal{R}$, and $\mathcal{J}$ are defined in terms of the ideals, the inclusion order among these ideals induces a partial order among the corresponding equivalence classes as;

$$L_a \leq L_b \iff S^1 a \subseteq S^1 b,$$

$$R_a \leq R_b \iff a S^1 \subseteq b S^1,$$

$$J_a \leq J_b \iff S^1 a S^1 \subseteq S^1 b S^1.$$

Thus, we may regard $S/\mathcal{L}$, $S/\mathcal{R}$, and $S/\mathcal{J}$ as posets.

Immediately from the above definitions, for $a \in S$ and $x, y \in S^1$,

$$L_{xa} \leq L_a, \quad R_{ax} \leq R_a.$$

We shall say that a semigroup S satisfies the conditions $\min_L$, $\min_R$, $\min_J$, accordingly as the minimum condition is satisfied by the POSets $S/\mathcal{L}$, $S/\mathcal{R}$, and $S/\mathcal{J}$.

Proposition 3.1.6. *If S is a semigroup satisfying $\min_L$ and $\min_R$, then $\mathcal{D} = \mathcal{J}$.*

Proof. Suppose that S satisfies $\min_{\mathcal{L}}$ and $\min_{\mathcal{R}}$. Then so also does S^1 , for S^1 has exactly the same principal left and right ideals as S , except (in the case where $S^1 \neq S$) for S^1 itself. So we may assume that S has an identity element.

Suppose now that $a \mathcal{J} b$, so that there exist $p, q, r, s \in S$ such that $paq = b$, $rb s = a$.

It follows that the set $X = \{x \in S : (\exists y \in S) xay = b\}$ is non-empty, and hence so also is the subset $\Lambda = \{L_x : x \in X\}$ of $S/\mathcal{L}$. The condition $\min_{\mathcal{L}}$ allows us to select a minimal element L_u in Λ , and we can then choose an element v in S such that $uav = b$.

Now $uruavsv = b$, and so $L_{uru} \in \Lambda$. Then, we have $L_{uru} \leq L_u$, and it then follows by the minimality of L_u in Λ that $L_u = L_{uru} \leq L_{ru} \leq L_u$.

Hence $ru \mathcal{L} u$. By Proposition 3.1.2 it follows that $ruav \mathcal{L} uav$, that is, $rb \mathcal{L} b$.

A similar argument using the condition $\min_{\mathcal{R}}$ establishes that $bs \mathcal{R} b$, and then by Proposition 3.1.2 we deduce that $rb s \mathcal{R} rb$. We thus have $a \mathcal{R} rb$, $rb \mathcal{L} b$, which implies $a \mathcal{R} \circ \mathcal{L} b$, and so $a \mathcal{D} b$ as required. $\square$

3.2 The structure of $\mathcal{D}$ -classes

Each $\mathcal{D}$ -class in a semigroup S is a union of $\mathcal{L}$ -classes and also a union of $\mathcal{R}$ -classes. The intersection of an $\mathcal{L}$ -class and an $\mathcal{R}$ -class is either empty or is an $\mathcal{H}$ -class. Hence, by the definition of $\mathcal{D}$,

$$a \mathcal{D} b \iff R_a \cap L_b \neq \emptyset \iff L_a \cap R_b \neq \emptyset.$$

We visualize a $\mathcal{D}$ -class in a table, from which each row represents an $\mathcal{R}$ -class, each column represents an $\mathcal{L}$ -class, and each cell represents an $\mathcal{H}$ -class, usually called as eggbox.

Remark 3.2.1. Let D be a $\mathcal{D}$ -class in a semigroup S , and let $a, b \in D$ are such that $a \mathcal{R} b$ (so a and b are in the same row of eggbox), then by definition, of $\mathcal{R}$ there exist $s, s' \in S^1$ such that

$$as = b, \quad bs' = a.$$

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The right translation $\rho_s : S \rightarrow S$ such that $x\rho_s = xs$, thus maps a to b . It, in fact, maps L_a to L_b ; for if $x \in L_a$ then $xs \mathcal{L} a$ as by proposition 3.1.2, and so $xs \in L_a s = L_b$. Similarly, $\rho_{s'}$ maps L_b to L_a .

Now if we consider the composition $\rho_s \rho_{s'} : L_a \rightarrow L_a$, we have for $x = ua \in L_a$

$$x\rho_s \rho_{s'} = uass' = ubs' = ua = x.$$

Thus, $\rho_s \rho_{s'}$ is identity map of L_a .

Similarly, $\rho_{s'} \rho_s$ is the identity map of L_b . Consequently, we deduce that $\rho|_{L_a}$ and $\rho|_{L_b}$ are mutually inverse bijections from L_a onto L_b and L_b onto L_a , respectively.

In fact, if $x \in L_a$, then $y = x\rho_s \in L_b$ has the property that $y = xs$ and $x = ys'$. Thus, $x \mathcal{R} y$ and so the map ρ_s is $\mathcal{R}$ -class preserving. It maps each $\mathcal{R}$ -class in L_a in a one-one manner onto the corresponding ($\mathcal{R}$ equivalent) $\mathcal{R}$ -class in L_b . Similar remark apply to $\rho_{s'}$.

To summarise, we have the following result:

Lemma 3.2.1 (Green's lemma). *Let a, b be $\mathcal{R}$ -equivalent elements in a semigroup S , and let $s, s' \in S^1$ be such that*

$$as = b, \quad bs' = a.$$

Then the right translations $\rho_s|_{L_a}$ and $\rho_{s'}|_{L_b}$ are mutually inverse $\mathcal{R}$ -class preserving bijections from L_a onto L_b and from L_b onto L_a , respectively.

Similarly, for a, b belonging to the same $\mathcal{L}$ -class, the left translations $\lambda_s|_{R_a}$ and $\lambda_{s'}|_{R_b}$ are mutually inverse bijections preserving $\mathcal{L}$ -classes from R_a onto R_b and from R_b onto R_a , respectively. We have the corresponding result as follows (also known as Green's Lemma):

Lemma 3.2.2. *Let a, b be $\mathcal{L}$ -equivalent elements in S , and let $t, t' \in S^1$ be such that*

$$ta = b, \quad t'b = a.$$

Then the left translations $\lambda_t|_{R_a}$ and $\lambda_{t'}|_{R_b}$ are mutually inverse $\mathcal{L}$ -class preserving bijections from R_a onto R_b and from R_b onto R_a , respectively.

The combined effect of these two lemmas is as follows;

Lemma 3.2.3. *If a, b are $\mathcal{D}$ -equivalent elements in a semigroup S , then $|H_a| = |H_b|$.*

Proof. Let c be such that $a \mathcal{R} c$ and $c \mathcal{L} b$, with $as = c$, $cs' = a$, $tc = b$, $t'b = c$.

Then by the preceding lemmas $\rho_s|_{H_a}$ is a bijection onto H_c and $\lambda_t|_{H_c}$ is a bijection onto H_b . Thus $\rho_s \lambda_t : x \mapsto txs$ is a bijection from H_a onto H_b (with inverse $\lambda_{t'} \rho_{s'} : y \mapsto t'ys'$), and it follows that $|H_a| = |H_b|$. $\square$

We have seen that if $a \mathcal{R} a$, then $x \mapsto xs$ is a bijection from H_a onto H_{as} . So, in particular, if $a \mathcal{H} a$ then $x \mapsto xs$ is a bijection of H_a onto itself. This, together with the dual argument, gives

Lemma 3.2.4. *Let x, y be elements of a semigroup S . If $xy \in H_x$ then $\rho_y|_{H_x}$ is a bijection of H_x onto itself. If $xy \in H_y$ then $\lambda_x|_{H_y}$ is a bijection of H_y onto itself.*

Then we have the following result, usually called Green's Theorem:

Theorem 3.2.5 (Green's Theorem). *If H is an $\mathcal{H}$ -class in a semigroup S then either $H^2 \cap H = \emptyset$ or $H^2 = H$ and H is a subgroup of S .*

Proof. Suppose $H^2 \cap H \neq \emptyset$, so that there exist a, b in H such that $ab \in H$. By the lemma, ρ_b and λ_a are bijections of H onto itself. Hence $hb \in H$ and $ah \in H$ for every h in H . Again by the lemma, it follows that λ_h and ρ_h are bijections of H onto itself. Hence $Hh = hH = H$ for every h in H . Certainly $H^2 = H$, and it follows from that H is a subgroup of S . $\square$

Corollary 3.2.6. *If e is an idempotent in a semigroup S , then H_e is a subgroup of S . No $\mathcal{H}$ -class in S can contain more than one idempotent.*

3.3 Regular $\mathcal{D}$ -classes

Definition 3.3.1. Let S be a semigroup and let $a \in S$. Then a is said to be *regular* if $\exists x \in S$ such that $axa = a$. The semigroup S is called *regular* if all of its elements are regular.

If a is a regular element with $axa = a$, and $b \mathcal{L} a$, then by definition, $\exists u, v \in S^1$ such that $ua = b$, $vb = a$, and so

$$b = ua = uaxa = bxa = b(xv)b.$$

Hence b is also regular.

Similarly, if a is regular with $axa = a$, and $b \mathcal{R} a$, then $\exists u, v \in S^1$ such that $au = b$, $av = a$, and so

$$b = au = axau = axb = b(vx)b.$$

Hence b is again regular.

Proposition 3.3.1. *If a is a regular element of a semigroup S , then every element of D_a is regular.*

Thus, if D is a $\mathcal{D}$ -class, then either all of the elements of D are regular or no element of D is regular.

If all of elements of D are regular then we call it regular; otherwise irregular.

Again, if e is an idempotent, then since $e = eee$, e is regular. It follows that if a $\mathcal{D}$ -class contains an idempotent then it is regular. The converse is also true: every regular $\mathcal{D}$ -class must contain atleast one idempotent. In fact, we have the following generalised result:

Proposition 3.3.2. *In a regular $\mathcal{D}$ -class, each $\mathcal{L}$ -class and each $\mathcal{R}$ -class contains an idempotent.*

Proof. If a is a member of a regular $\mathcal{D}$ -class, and $xax = a$, then xa is idempotent and $xa \mathcal{R} a$, while ax is idempotent and $ax \mathcal{L} a$. □

Proposition 3.3.3. *Every idempotent e in a semigroup S is a right identity for L_e and a left identity for R_e .*

Proof. If $a \mathcal{L} e$, then $a = xe$ for some $x \in S^1$, and so

$$ae = (xe)e = xe^2 = xe = a.$$

Similarly, if $b \mathcal{R} e$, then $b = ex$ for some $x \in S^1$, and so

$$eb = e(ex) = e^2x = ex = b.$$

□

Definition 3.3.2 (Inverse element). Let S be a semigroup and $a \in S$. We say that a' is an *inverse* of a if $aa'a = a$ and $a'aa' = a'$.

Remark 3.3.3. Every element with an inverse is regular. In fact, every regular element has an inverse. To see this, let a be a regular element with $axa = a$, then define $a' = xax$. So we have a' is an inverse of a . Indeed, $aa'a = a(xax)a = (axa)xa = axa = a$, and $a'aa' = (xax)a(xax) = x(axa)xax = x(axa)x = xax = a'$.

Remark 3.3.4. An element may have more than one inverse. For example, in a rectangular band every element is an inverse of every other element.

We shall denote the set of inverses of an element a by $V(a)$.

Now, notice that if a is an element of a regular $\mathcal{D}$ -class D , then every inverse a' of a must lie in D , as $a\mathcal{R}aa'$ and $a\mathcal{L}a'a$.

	L_a	$L_{a'}$
R_a	a	aa'
$R_{a'}$	$a'a$	a'

We have proved part of the following theorem:

Theorem 3.3.4. *Let a' be an element of a regular $\mathcal{D}$ -class D in a semigroup S .*

- (1) *If $a' \in V(a)$, then $a' \in D$ and the two $\mathcal{H}$ -classes $R_a \cap L_{a'}$, $L_a \cap R_{a'}$ contain the idempotents aa' and $a'a$, respectively.*
- (2) *If $b \in D$ is such that $R_a \cap L_b$ and $L_a \cap R_b$ contain idempotents e and f , respectively, then H_b contains an inverse a^* of a such that $aa^* = e$, $a^*a = f$.*
- (3) *No $\mathcal{H}$ -class contains more than one inverse of a .*

Proof. It remains to establish the second and third parts.

	L_a	L_b
R_a	a	e
R_b	f	a^*, b

To prove (2), notice that from $a\mathcal{R}e$ it follows by Proposition 3.3.3 that $ea = a$. Similarly, from $a\mathcal{L}f$ it follows that $af = a$. Again from $a\mathcal{R}e$ it follows that there exists x in S^1 such that $ax = e$. Let $a^* = fxe$. Then

$$\begin{aligned} aa^*a &= (af)x(ea) = axa = ea = a, \\ a^*aa^* &= fx(eaf)xe = fx(ax)e = fxe^2 = fxe = a^*, \end{aligned}$$

and so $a^* \in V(a)$. Also

$$aa^* = (af)xe = (ax)e = e^2 = e.$$

Further, since $a\mathcal{L}f$, there exists y in S^1 such that $ya = f$. Hence

$$a^*a = fxea = fxa = yaxa = yea = ya = f.$$

It now follows easily that

$$a^* \in L_e \cap L_f = L_b \cap R_b = H_b.$$

To prove part (3), suppose that both a' and a^* are inverses of a inside the single $\mathcal{H}$ -class H_b . It follows that both aa' and aa^* are idempotents in the $\mathcal{H}$ -class $R_a \cap L_b$. Hence $aa' = aa^*$ by Corollary 3.2.6. Similarly,

$$a'a = a^*a,$$

and it now follows that

$$a^* = a^*aa^* = a^*aa' = a'aa' = a'.$$

□

We have the following easy consequence of the above theorem:

Proposition 3.3.5. *Let e, f be idempotents in a semigroup S . Then $(e, f) \in \mathcal{D}$ if and only if there exists $a \in S$ and an inverse a' of a such that $aa' = e, a'a = f$.*

Proof. Suppose first that $(e, f) \in \mathcal{D}$. Then e and f are members of the same regular $\mathcal{D}$ -class.

	L_f	L_e
R_e	a	e
R_f	f	a'

Let $a \in R_e \cap L_f$. Then by Theorem 3.3.4(2) there is an inverse a' of a in $R_f \cap L_e$ such that

$$aa' = e, \quad a'a = f.$$

Conversely, if there exist mutually inverse elements a, a' such that

$$aa' = e, \quad a'a = f,$$

then from Theorem 3.3.4(1) it follows that $e\mathcal{R}a, a\mathcal{L}f$. Hence $e\mathcal{D}f$ as required. $\square$

Proposition 3.3.6. *If H and K are two group $\mathcal{H}$ -classes in the same regular $\mathcal{D}$ -class, then H and K are isomorphic.*

Proof. The method of proof is essentially that used in the proof of Lemma 3.2.3, with a careful (yet obvious) choice of translation maps. Suppose that $H = H_e$ and $K = H_f$, where e, f are idempotents and are, respectively, the identities of the groups H_e and H_f .

	L_e	L_f
R_e	e	a
R_f	a'	f

Let $a \in R_e \cap L_f$ (which is non-empty since $e\mathcal{D}f$ by assumption); then by Theorem 3.3.4 there is a unique inverse a' of a in $L_e \cap R_f$, and we have

$$aa' = e, \quad a'a = f, \quad ea = af = a, \quad a'e = fa' = a'.$$

By Green's Lemmas (3.2.1 and 3.2.2) it follows that $\rho_a|_{H_e}$ is a bijection from H_e onto H_a and $\lambda_{a'}|_{H_a}$ is a bijection from H_a onto H_f . Thus the map $\phi = \rho_a\lambda_{a'}$ given by

$$x\phi = a'xa \quad (x \in H_e)$$

is a bijection from H_e onto H_f , with inverse given by

$$y\phi^{-1} = aya' \quad (y \in H_f).$$

The bijection ϕ is even an isomorphism, for if $x_1, x_2 \in H_e$ then

$$(x_1\phi)(x_2\phi) = a'x_1aa'x_2a = a'x_1ex_2a = a'(x_1x_2)a = (x_1x_2)\phi,$$

since e is the identity element of the group H_e . $\square$

We have a final result to conclude the section:

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Proposition 3.3.7. *Let $a, b \in D$, where D is a $\mathcal{D}$ -class. Then $ab \in R_a \cap L_b$ if and only if $L_a \cap R_b$ contains an idempotent.*

Proof. Suppose first that $ab \in R_a \cap L_b$. Then there exists c such that $abc = a$, and by Green's Lemma 3.2.1 the right translation

$$\rho_c : x \mapsto xc$$

maps H_b onto $L_a \cap R_b$. Thus in particular $bc \in L_a \cap R_b$. The translation

$$\rho_b : y \mapsto yb$$

maps $L_a \cap R_b$ onto H_b , and is the inverse of ρ_c , and it then follows that bc is idempotent, since

$$(bc)^2 = b\rho_c\rho_b\rho_c = b\rho_c = bc.$$

Conversely, suppose that $L_a \cap R_b$ contains an idempotent e . Then $eb = b$ by Proposition 3.3.3, and hence the translation $x \mapsto xb$ maps H_a onto $R_a \cap L_b$. In particular, $ab \in R_a \cap L_b$. $\square$